Biometric Authentication Mechanisms

***Abstract*—Modern authentication commonly uses *biometric credentials* such as fingerprints and faces: a sensor evaluates each scan and accepts if its score exceeds a *match threshold*. A large body of work improves sensor accuracy but sets this threshold heuristically. Recent work studies *authentication mechanisms* that combine credentials to maximize the probability of accepting the user and rejecting the attacker. But it treats every credential as *atomic*: presented correctly or not, like a password.**

We model authentication mechanisms with both biometric and atomic credentials. For a single-attempt mechanism that uses a biometric credential alone, threshold optimization yields only marginal improvements. However, when biometrics are combined with other credentials or used across multiple attempts, threshold optimization becomes consequential. We prove that a mechanism combining an atomic credential with an optimal-threshold biometric one is better than a mechanism using only the atomic credential; this holds even if the biometric credential is extremely inaccurate. Next, we consider mechanisms allowing multiple attempts, as fingerprint readers commonly do. We find that each attempt's threshold should be different. Using public biometric datasets, we find that optimal thresholds can increase the success probability of multiple-attempt mechanisms by 72 percentage points compared to the heuristic threshold.

Yet our survey of commercial sensors finds that almost none allow setting a specific threshold. Based on our results, manufacturers should change their interfaces so integrators can deploy optimal authentication mechanisms.

1. Introduction

Biometric credentials are increasingly adopted for access control, as they appear in 81% of smartphones and 70% of financial banking services, mainly through face or fingerprint verification [1], [2]. They rely on unique physical properties that make them difficult to forge [3]. Biometric credentials are evaluated by a sensor that obtains a *sample*. It calculates a *similarity score* between that sample and a *reference* stored in advance, and outputs *accept* if the score is above a *match threshold* or *reject* otherwise. This is in contrast to *atomic credentials*, which are either presented correctly or not. Those are evaluated according to a perfect match criterion.

Previous work (§2) analyzed biometric credentials in isolation, primarily focusing on reducing the *False Acceptance Rate* and *False Rejection Rate* (*FAR*, *FRR*) [4], [5], [6]. Sarkar et al. [7] suggest to set the threshold heuristically to obtain an Equal Error Rate (*EER*, where $FAR = FRR$). But these works do not analyze the integration of biometric credentials into an authentication mechanism. Other works analyze authentication mechanisms as predicates of credential possession [8], [9], [10], [11], but treat all credentials as atomic.

In this work, we study how to design *optimal mechanisms*, mechanisms with the maximum probability of success given the credentials and their probabilities, including biometric credentials. First, we present a refined definition of the authentication problem with biometric credentials (§3). The system comprises an authentication mechanism and two principals: a *user* and an *attacker*. Each principal interacts with the mechanism by presenting credentials.

Biometric sensors aim to assign high scores to samples from legitimate users and low scores to samples from attackers. The scores obtained by comparing legitimate users' samples against the reference follow the *genuine distribution*. The scores obtained by comparing attackers' samples against the reference follow the *impostor distribution*. Consequently, the genuine distribution should be shifted toward higher matching scores compared to the impostor distribution, as shown in Figure 1. These distributions establish two fault probabilities that depend on the threshold. The *FAR* is the probability that the impostor score value is above the threshold (red volume) and the *FRR* is the probability that the legitimate user's score value is below the threshold (blue volume).

To identify an optimal mechanism, we derive closed-form expressions for the mechanism's success probability, using first and second order approximations of both principals' biometric score distributions. We start by optimizing a single biometric credential mechanism (§5). In a single credential mechanism, the mechanism success probability reduces to the probability that the credential is *safe*, i.e., only to the user can use the credential. We identify the optimal threshold of the biometric credential and show that, in general, the optimal threshold differs from the commonly used *EER* threshold.

Next, we extend our analysis to a mechanism that combines two credentials (§6). A mechanism that uses two biometric credentials appears in manufacturing industry and in financial services [12]. We evaluate a mechanism comprising two biometric credentials by considering first and second order approximations of the biometric credential distributions. We then use a numerical optimization algorithm to find the optimal thresholds for each biometric credential. The optimal thresholds of the biometric credentials in such a mechanism are not, in general, the same as the optimal

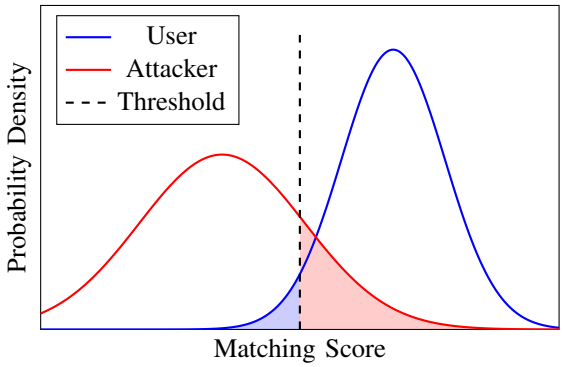


Figure 1: User and Attacker score distributions with *FAR* and *FRR* areas appear in red and blue respectively.

thresholds if used individually and differs from the *EER*. This difference arises as we optimize each credential with respect to the entire mechanism. To demonstrate the potential impact, we present a synthetic experiment of two sampled score distributions and observe a significant reduction in the failure rate, when comparing using the *EER* and the optimal threshold.

We then turn to analyze mechanisms that use both a biometric credential and an atomic credential. Such mechanisms are prevalent in real-world authentication systems: for example, Google [13] assumes (1) possession of a trusted device like a phone or computer (Atomic credential) and (2) a biometric credential. We derive closed-form expressions for the mechanism's success probability. Our symbolic analysis reveals that the relative influence of the atomic and biometric credentials on the mechanism's success probability, depends on their individual properties. In particular, as one credential becomes increasingly reliable, the mechanism success probability depends mainly on that credential. Consequently, changes in the less reliable credential have a smaller effect on the mechanism success probability.

Motivated by this observation, we identify the conditions under which the integration of a biometric credential with an atomic credential will always enhance the mechanism's success probability (§7). First, we show that when an atomic credential can only be *safe* or *stolen*, it is not worthwhile to use both the atomic and biometric credentials. This is because a mechanism that uses an atomic credential alone succeeds whenever the credential is safe. However, adding a biometric credential may reject the user even if the atomic credential is safe, without improving the mechanism's success when the atomic credential is stolen. Next, we consider atomic credentials that can also be *leaked*, where both the user and attacker can use the credential, or *lost*, where neither can use it. We introduce a necessary and sufficient condition we call *asymptotic separability*, of a biometric credential: the *FAR* converges to zero for matching scores above a given threshold, or the *FRR* converges to zero for matching scores below it. This assumption is reasonable since biometric sensors aim to distinguish between legitimate users and attackers. Suppose the atomic credential has non-zero probabilities for the states of *safe* and either *loss* or *leak* and any probability of *stolen*. We prove that every *asymptotically separable* biometric credential that is optimally integrated with such an atomic credential, yields a mechanism whose success probability dominates that of the atomic credential mechanism.

Biometric mechanisms typically use the same fixed hueristic threshold [14]. Motivated by this behaviour, we extend our analysis to a mechanism that uses a biometric credential only and allows multiple authentication attempts by each principal (§8). In practice, biometric real-world authentication mechanisms allow principals to retry authentication following a failed attempt. Examples appear in Apple and Samsung devices [15], [16]. However, while additional sampling opportunities increase the legitimate user's success probability, they also provide attackers with additional opportunities to authenticate successfully. Under a bounded number of attempts mechanism, the mechanism succeeds if the legitimate user authenticates at least once while the attacker fails across all attempts. We then derive closed-form expressions for the mechanism's success probability and optimize the match thresholds for each number of attempts. Our analysis shows that the optimal threshold increases with the number of attempts. However, system designers may seek to limit the expected number of attempts required for the legitimate user to successfully authenticate. We therefore propose a different approach in which we don't bound the number of attempts, but maximize the success probability subject to a bound on the mean number of attempts. Conversely, we minimize the mean number of attempts subject to a required success probability. These optimization problems reveal the tradeoff between security and usability in the authentication system.

We quantitatively evaluate our framework (§**??**) using public biometric data [17]. We employ a standard fingerprint verification algorithm [18] to evaluate the resulting matching scores of fingerprints from multiple users. We produce the user distribution by matching fingerprints from the same user, resulting in high matching scores and the attacker distribution by matching fingerprints from different users. Consequently, the resulting attacker distribution concentrates around low score values. As a result, the user and the attacker distributions share a small overlapping area, resulting in a very high success probability. Thus, no significant improvement occurs when comparing the mechanism's success under the *EER* threshold and the optimal threshold. These results suggest that the *EER* threshold is sufficient for this attack scenario.

However, the advantage of threhsold optimization becomes apparent when attackers are presenting forged biometric samples. Thus, we conduct a spoofing attack evaluation using the LivDet database [17], [19], [20], [21], which includes both genuine fingerprint scans and corresponding spoofing samples. We match each user's spoofing samples against their geniune samples to obtain the imposter score distribution. These spoof samples produce matching scores that closely resemble those of the legitimate user, increasing the biometric credential fault probabilities. Thus, threshold

optimization becomes necessarry in a spoofing attack scenario. We show that integrating the biometric credential into a mechanism that uses an atomic credential and optimizing the threshold, reduces the mechanism's failure probability by 20-72% compared to its failure probability under the *EER* threshold. Furthermore, the results show a reduction of up to 21% in the mechanism's failure probability compared to a mechanism based solely on an atomic credential, demonstrating the benefit of taking advantage of the biometric credential.

Our previous evaluation demonstrates that optimizing the match threshold significantly improves the mechanism's success probability under spoofing attacks. However, in practice, an attacker may perform multiple authentication attempts using spoofed fingerprints, potentially increasing his probability of successfully authenticating. We therefore extend our evaluation to examine whether threshold optimization remains beneficial under multiple authentication attempts. Using the same user and attacker distributions, we compare the mechanism success probability when the thresholds optimized jointly across attempts with two baselines: reusing the *EER* threshold and reusing the optimal single attempt threshold. Our results show that the mechanism success probability is lower by up to 75% on both baselines than jointly optimizing the thresholds. Moreover, the optimal thresholds value increase with the number of attempts, as predicted by our theoretical analysis.

To assess whether commercial biometric sensors support threshold optimization, we review their specifications $(\S9)$. We examine 32 biometric device families from 21 manufacturers, covering different modalities, including fingerprints, face, iris, and palm recognition. Our findings reveal that only two device families permit tuning a numeric match threshold, while the others expose target error rates, discrete security levels, or no threshold control at all. However, setting a threshold is necessary but not sufficient for optimization: the integrator must also know *FAR* and *FRR* functions to choose a threhsold. Applying this criterion reveals that no family publishes these functions. Consequently, no surveyed sensor supports threshold optimization, limiting the deployment of optimal authentication mechanisms.

In summary $(\S10)$, our contributions are both theoretical and practical: (1) Formalization of the success probability function for authentication mechanisms using biometric and atomic credentials; (2) Symbolic analysis of the optimal threshold for a mechanism using a biometric credential; (3) Identification of the conditions under which adding a biometric credential improves the mechanism's success probability; (4) Proof that any biometric credential satisfying these conditions improves the mechanism's success probability; (5) Derivation of the mechanism's success probability under multiple authentication attempts; (6) Formulation of constrained optimization problems balancing the mechanism's success probability and the mean number of authentication attempts; (7) Empirical evaluation of multiple authentication attempts under spoofing attacks using real biometric data; (8) Practical implication that biometric sensor vendors should enable threshold configuration to improve authentication security.

2. Related Work

Authentication is a constant active and prominent research area, with prior work broadly categorized into two main aspects: credentials and mechanisms. Credentials are commonly manifested as passwords [22], One-Time-Password (OTP) [23], physical devices [24] or biometric features such as fingerprints, face recognition or iris [25].

Extensive work has evaluated and improved the accuracy of biometric systems [4], [6], [17], [19], [20], [21], [26]. Prabhakar et al. [5] examine the tradeoff between two key error metrics of a biometric system, namely False Acceptance Rate (*FAR*) and False Rejection Rate (*FRR*). Sarkar et al. [7] define the term Equal Error Rate (*EER*), the threshold where *FAR* and *FRR* are equal. They heuristically suggest it as the threshold for biometric credentials. However, these works do not define a success criterion for authentication mechanisms, nor do they explore how the presence of additional credentials can influence the optimal matching threshold of the biometric system. Hong et al. [27] integrate fingerprint and face recognition, each one operates independently, using an *AND mechanism*. They design a decision fusion protocol that makes a decision considering the *FAR* error metric only. However, they do not explore how one biometric credential affects the other for optimal usage, nor do they analyze the integration of biometric credentials along with other credential types within an authentication mechanism.

An authentication mechanism is a function that evaluates a set of credentials provided by a principal [11] and output accept or reject [28], [29]. Recent work has explored authentication mechanism defined as predicate of credential possession. Maram et al. [8] study interactive authentication mechanisms, where the user and attacker interact with the authentication mechanism. Mouallem et al. [9] compare mechanisms' success and aim to identify mechanisms with maximal success when composed of a large number of homogeneous credentials. Eyal [10] analyzes mechanisms modeled as Boolean functions of the credentials. However, these works are partially applicable to our setting. While we adopt their mechanism's success definition and the way they define the credential possession states, they model biometric credentials as atomic credentials neglecting the tuneable threshold.

3. Model

We formalize the authentication problem by specifying an execution and the system participants $(\S3.1)$, credential types and their characteritics $(\S3.2)$, and the authentication mechanism and its success $(\S3.3)$.

3.1. System Participants

The system comprises an authentication mechanism M and two principals: a user U and an attacker A. In an execution E, a principal P *presents* a credential c_i to the

mechanism by calling a random function *present credentials* denoted by $present_P(i)$. We abstract away the concrete implementation of a credential and model only its presentation. The value of the *presentation function* determines whether a credential is available to a principal. The entry $\sigma_i^P = 1$ means that P successfully presented credential c_i, and $\sigma_i^P = 0$ means that it is not. At the beginning of an execution E, the system is parameterized such by an *allocation function*. A credential is successfully presented with probability randomized by the allocation function denoted by $allocate_P(i)$.

Definition 1. *(Availability Vector) An* availability vector *specifies, for each credential, whether it is successfully presented by a principal P or not. Denote by σ^U and σ^A the availability vectors of the user and of the attacker, respectively.*

$$\sigma_i^P = present_P(i)$$

The outcome of the presentation function determines whether $\sigma_i^P = 1$ or $\sigma_i^P = 0$. A scenario σ is a tuple of availability vectors (σ^U, σ^A).

3.2. Credentials

A credential may correspond to secret data, such as a password or one time password (OTP); a physical object, such as a phone or security key; or a biometric property, such as a fingerprint or face. We model these credentials as either *atomic* or *biometric*. Each credential can be either: *Safe*: Only available to the user, *Lost*: Not available to either principal, *Leaked*: Available to both principals, or *Stolen*: Only available to the attacker.

3.2.1. Passwords, OTPs and security keys are credential types evaluated according to a perfect match criterion, namely, atomic credentials.

Definition 2. *(Atomic Credential) For an atomic credential the allocation function $allocate_P(i)$ randomly draws the state probabilities of the credential, namely P_{safe}, P_{loss}, P_{leak} and P_{stolen}.*

When a principal P presents an atomic credential c_i, the presentation function ouputs 1 with probability $P_{\text{safe}} + P_{\text{leak}}$ when allocated to the user and 0 with probability $P_{\text{leak}} + P_{\text{stolen}}$ when allocated to the attacker.

3.2.2. Alternatively, a credential may correspond to a biometric property, such as a fingerprint or face. The biometric sensor evaluates a measurement of the biometric property and produces a score s. The user distribution $f_U(s)$ denotes the score distribution of samples from the user, whereas the attacker distribution $f_A(s)$ denotes the score distribution of samples from the attacker.

Definition 3. *(Biometric Credential) For a biometric credential, the user score distribution $f_U(s)$, the attacker score distribution $f_A(s)$, and the match threshold t define the presentation function $present_P(i)$. For each principal $P \in \{U, A\}$, the allocation function $allocate_P(i)$ draws a random score s from the distribution $f_P(s)$. Thus*

$$s = allocate_P(i)$$

$$present_P(i) = \begin{cases} 1 & s \geq t \\ 0 & otherwise \end{cases}$$

We denote by σ_i the state of credential c_i in scenario σ. The user and attacker distributions establish two error metrics:(1) *FAR*, the false acceptance rate, is the probability that the sensor outputs *accept* for an attacker sample. We calculate it by integrating the attacker distribution from the match threshold to the maximum score, thus

$$FAR(t) = \int_t^{\infty} f_A[s]ds \tag{1}$$

(2) *FRR*, the false rejection rate, is the probability that the sensor outputs *reject* for a user sample. We calculate it by integrating the user distribution from the minimum score to the match threshold, thus

$$FRR(t) = \int_{-\infty}^{t} f_U[s]ds \tag{2}$$

We use *FAR* and *FRR* to find the state probabilities of the biometric credential. The user has access to the biometric credential when the output of the biometric sensor is *accept*, which occurs with probability $1 - FRR(t)$. The attacker has access to the biometric credential when the output of the biometric sensor is *accept*, which occurs with probability $FAR(t)$. Thus, the probabilities of the four states are

$$P_{\text{loss}} = FRR(t) \cdot (1 - FAR(t)),$$

$$P_{\text{leak}} = FAR(t) \cdot (1 - FRR(t)),$$

$$P_{\text{stolen}} = FRR(t) \cdot FAR(t), \text{ and}$$

$$P_{\text{safe}} = 1 - P_{\text{loss}} - P_{\text{leak}} - P_{\text{stolen}} = (1 - FAR(t)) \cdot (1 - FRR(t)). \tag{3}$$

3.3. Mechanism

An authentication mechanism M receives the *evaluation* of the presentation function $present_P(i)$ for each principal P and outputs a *decision*.

Definition 4. *(Authentication Mechanism) An authentication mechanism M is a predicate that takes as input:*

- *σ^P the availability vector of a principal P in scenario σ (perhaps an empty vector, represented by $\varnothing$).*

and outputs either 1, meaning that the mechanism accepts the principal or 0, meaning that the mechanism rejects the principal.

We are now ready to define the mechanism's success. The mechanism is successful if it grants access to the user only.

Definition 5. *(Mechanism Success) A mechanism M is* successful *in a scenario σ if for a principal P and its availability vector σ^P, the authentication mechanism decision* $\mathrm{decide}_M(\sigma^P)$ *equals 1 for the user only over an execution E, that is*

$$\mathrm{decide}_M(\sigma^U) = 1 \wedge \mathrm{decide}_M(\sigma^A) = 0.$$

Otherwise, the mechanism is unsuccessful.

The mechanism's *success probability* is thus

$$P_{\text{Success}}(M) = \Pr\left[\mathrm{decide}_M(\sigma^U) = 1 \wedge \mathrm{decide}_M(\sigma^A) = 0\right].$$

For example, in an *AND mechanism* [10], the mechanism grants access to a principal if and only if all credentials are available to the principal; thus, $\mathrm{decide}_M(\sigma^P) = 1$ if σ_i^P equals 1 for every credential c_i. In an *OR mechanism*, the mechanism grants access to a principal if at least one credential is available to the principal; thus, $\mathrm{decide}_M(\sigma^P) = 1$ if σ_i^P equals 1 for at least one credential c_i.

4. Real Data and Approximations

We evaluate our model using the public real-world *LivDet2015* dataset of fingerprints [17], [19], [20], [21]. This dataset contains multiple fingerprint scans of each user, acquired using different types of biometric sensors, as well as spoof samples produced from each user's fingerprints. We divide each user's scans into a reference set, representing reference scans stored in advance, and a candidate set, representing scans obtained by the sensor. To score the fingerprint scan images, we use the standard fingerprint verification algorithm *SourceAFIS* [18]. This algorithm compares the relative positions and orientations of fingerprint ridge endings and bifurcations across two images. We compute a similarity score for each candidate scan by comparing it against every reference image, and retaining the maximum score as the candidate's final score [6], [27], [30]. Following the evaluation procedures of Jain et al. [4], Cappelli et al. [6] and Slobodan et al. [31], we produce histograms of genuine and imposter comparisons, to obtain the empirical score distributions of the users and attackers, respectively, under two baseline attack scenarios.

In the first scenario, an attacker authenticates using their own fingerprints. The genuine comparisons involve comparisons between all the user's candidate images and the user's reference images. By contrast, the impostor comparisons involve comparisons between all the user's reference images and the scans of the other users in the database. With an *EER* of approximately 3.8%, the algorithm's performance is representative of state-of-the-art fingerprint verification [6], [26] and the best results achieved in the *LivDet2015* competition [17]. In the second scenario, an attacker creates a fake fingerprint using spoofing materials, produced by *Gelatine*, *Woodglue*, *Latex* and *Ecoflex*. We match these spoof samples against the user's reference images to obtain the impostor score distribution.

To symbolically analyze different mechanisms, we approximate the empirical score distributions using theoretical distributions. Evaluating different mechanisms requires the biometric credential's fault probabilities which are determined by the *FAR* and *FRR* values (Equation 3). We evaluate the accuracy of the theoretical models introduced in (§5) by fitting uniform, quadratic, and normal distributions to the genuine and impostor score histograms in both attack scenarios. For each fitted distribution, we calculate the Kolmogorov-Smirnov (KS) statistic [32]. This statistic measures the maximum absolute difference between the empirical and fitted cumulative distribution functions, which correspond to the *FRR* and *FAR* curves. Figure 2 presents the resulting *FAR* and *FRR* curves alongside the empirical values for both scenarios. The KS statistics range from 0.04 to 0.07 for the genuine scores and are approximately 0.17 for the impostor scores, providing empirical support for using our theoretical models as approximations of the observed score distributions.

5. Mechanisms

We explore optimal mechanism success probability with a single biometric credential. We start with first (§5.1) and second (§5.2) order approximations.

5.1. Uniformly Distributed Biometric Credentials

We consider uniform distribution as a first-order approximation of the biometric credential distributions. Denote by u_1 *and* u_2 the user's PDF bounds and by a_1 *and* a_2 the attacker's PDF bounds as shown in Figure 3. The probability distribution functions for the user and the attacker, f_U and f_A are thus:

$$f_U(s) = \frac{1}{u_2 - u_1} \quad \text{for } u_1 \le s \le u_2 \text{ and}$$
$$f_A(s) = \frac{1}{a_2 - a_1} \quad \text{for } a_1 \le s \le a_2.$$

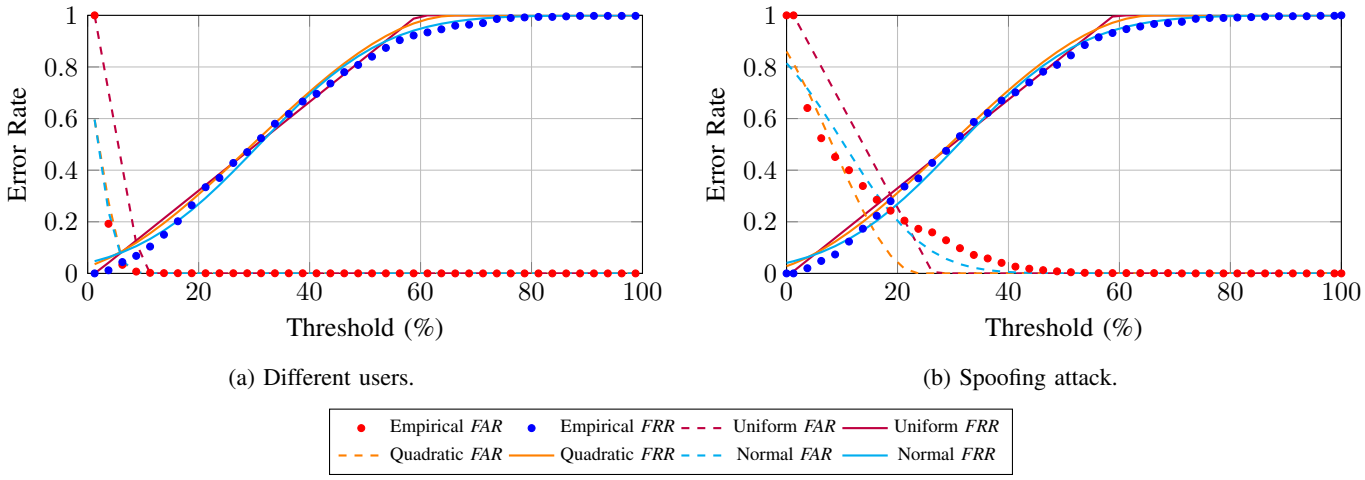
The error metrics *FAR* and *FRR* are both functions of the match threshold t and given by Equations 1 and 2. Thus:

$$\mathit{FAR}(t) = \int_t^{a_2} \frac{1}{a_2 - a_1} ds = \begin{cases} 1 & \text{if } t \le a_1 \\ \frac{a_2 - t}{a_2 - a_1} & \text{if } a_1 < t \le a_2 \\ 0 & \text{if } t > a_2 \end{cases}$$

and

$$\mathit{FRR}(t) = \int_{u_1}^{t} \frac{1}{u_2 - u_1} ds = \begin{cases} 0 & \text{if } t \le u_1 \\ \frac{t - u_1}{u_2 - u_1} & \text{if } u_1 < t \le u_2 \\ 1 & \text{if } t > u_2 \end{cases}$$

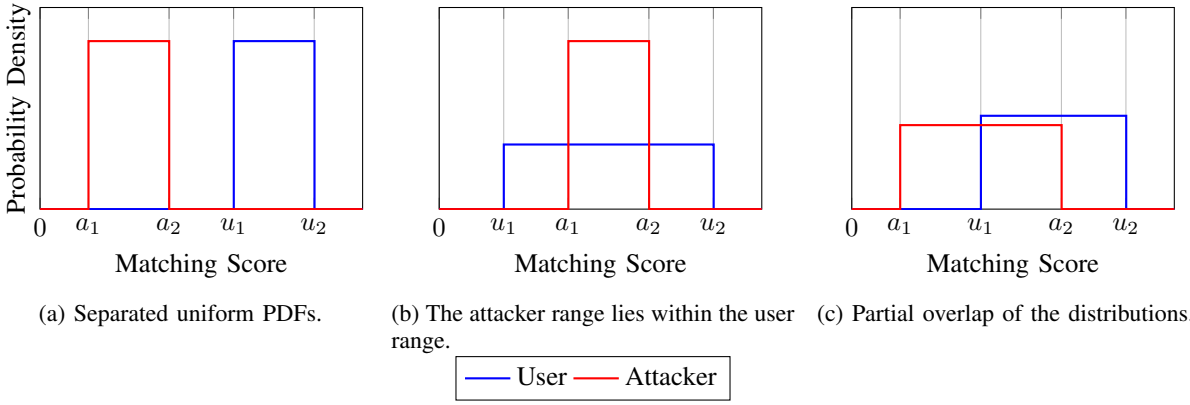
For a biometric credentials to be useful, users are expected to produce higher matching scores than attackers. Otherwise, the attacker would be more likely to pro-



(a) Different users.

(b) Spoofing attack.

Figure 2: Empirical *FAR* and *FRR* as functions of the threshold, compared with the uniform, quadratic and normal distribution functions obtained from the KS fits, under both attack scenarios.



(a) Separated uniform PDFs.

(b) The attacker range lies within the user range.

(c) Partial overlap of the distributions.

Figure 3: Quantitative illustration of three distinct cases of uniform attacker and user score distributions. In case (1), the distributions are separated. Case (2) illustrate the case where an attacker is fully contained in the user distribution, and (3) illustrate two overlapping configurations.

duce scores that lead to successful authentication, rendering the biometric credential ineffective. Thus, we assume that $a_2 < u_2$. There are three distinct cases:

(1) $a_2 < u_1$ (Figure 3a): In this case, the attacker and user distributions are *separated*. For a mechanism with a single credential M^{single}, the mechanism's success probability is simply the probability that the credential is safe

$$P_{\text{success}}(M^{\text{single}}) = P_{\text{safe}} \tag{4}$$

Any threshold $a_2 \leq t \leq u_1$ yields zero *FAR* and *FRR*, and therefore achieves a mehcanism success probability of 1.

(2) $u_1 < a_1 < a_2 < u_2$ (Figure 3b): The attacker distribution bounds lies within the range of the user distribution bounds. The success probability of the biometric credential is given by Equation 3. The *FAR* is one for any $t \leq a_1$ and zero for any $t \geq a_2$. Thus:

$$P_{\text{success}}(t) = \begin{cases} 0, & t \leq a_1 \\ (1 - \frac{a_2 - t}{a_2 - a_1}) \cdot (1 - \frac{t - u_1}{u_2 - u_1}), & a_1 < t \leq a_2 \\ 1 - \frac{t - u_1}{u_2 - u_1}, & a_2 < t \leq u_2 \\ 0, & t > u_2 \end{cases}$$

(3) $a_1 < u_1 < a_2 < u_2$ (Figure 3c): The user and attacker distributions partially overlap.

We maintain the same approach as the case (2) to calculate the probability of success. The *FAR* is one for any $t \leq a_1$ and zero for any $t \geq a_2$, and the *FRR* is zero for any $t \leq u_1$ and one for any $t \geq u_2$. Thus, in this case,

we obtain five regions of the success probability function:

$$P_{\text{success}}(t) = \begin{cases} 0, & t \le a_1 \\ 1 - \frac{a_2 - t}{a_2 - a_1}, & a_1 < t \le u_1 \\ (1 - \frac{a_2 - t}{a_2 - a_1}) \cdot (1 - \frac{t - u_1}{u_2 - u_1}), & u_1 < t \le a_2 \\ 1 - \frac{t - u_1}{u_2 - u_1}, & a_2 < t \le u_2 \\ 0, & t > u_2. \end{cases}$$

We analyze the probability of success in each interval. First, we show that $P_{\text{success}}(t)$ is continuous. Since $P_{\text{success}}(t)$ is continuous on each region of its piecewise definition, it remains only to check the boundary points. At each boundary point, the left-hand and right-hand limits of P_{success} coincide with P_{success} itself; therefore, P_{success} is continuous on its entire domain. Then, we find the optimal threshold that maximizes the mechanism's success probability in each region. The second and third regions of case (2) are equivalent to the third and fourth regions of case (3), thus we analyze them together. In case (3), the success probability function is linearly increasing in $a_1 \le t \le u_1$, and therefore its maximum is achieved at the right boundary, $t = u_1$. In both cases, the success probability is linearly decreasing on $a_2 \le t \le u_2$, and therefore its maximum is achieved at the left boundary, $t = a_2$. Since $t = u_1$ and $t = a_2$ are the boundary points of the region where *FAR* and *FRR* are strictly between zero and one, the maxima of the adjacent regions are already included in this region. Therefore, it is sufficient to analyze the region where $u_1 \le t \le a_2$ in case (3) or $a_1 \le t \le a_2$ in case (2), Figure 4 illustrates this analysis. Thus, we find the global maximum by differentiating the success probability function in this region and comparing the resulting critical point with the boundary values.

We differentiate the success function and set it to zero. We find that at: $t = \frac{a_1+u_2}{2}$ we get $\frac{d}{dt}P_{\text{success}}(t) = 0$ with $\frac{d}{dt}P_{\text{success}}(t) > 0$, for every $t \in (u_1, \frac{a_1+u_2}{2})$ *and* $\frac{d}{dt}P_{\text{success}}(t) < 0$, for every $t \in (\frac{a_1+u_2}{2}, a_2)$. Thus, the maximum point is at $t_{\text{Max}} = \frac{a_1+u_2}{2}$ if $\frac{a_1+u_2}{2} \in [u_1, a_2]$ in case (3) and $\frac{a_1+u_2}{2} \in [a_1, a_2]$ in case (2).

For both cases, the probability of success at the optimal point, if the condition is satisfied, is

$$P_{\text{success}}(t_{\text{opt}}) = \frac{(a_1 - u_2)^2}{4(a_1 - a_2)(u_1 - u_2)}.$$

For the overlap case, if the condition is not satisfied, the maximum point is at the boundary of the interval, which is either $t = u_1$ or $t = a_2$. Thus

$$P_{\text{success}}(t_{\text{opt}}) = \max\left\{\frac{u_1 - a_1}{a_2 - a_1}, \frac{u_2 - a_2}{u_2 - u_1}\right\}.$$

For the case where the attacker range lies within the user's range, if the condition is not satisfied the maximum point is at the lower boundary of the third interval, where $t = a_2$. Thus

$$P_{\text{success}}(t_{\text{opt}}) = \frac{u_2 - a_2}{u_2 - u_1}.$$

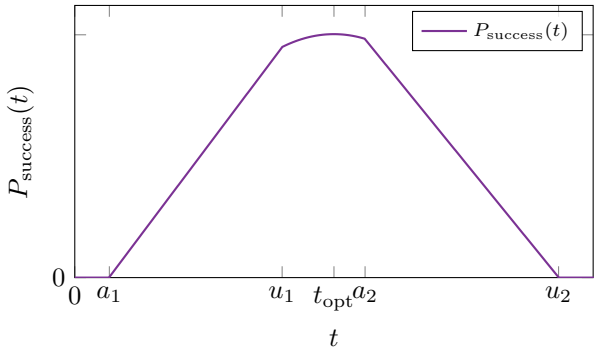


Figure 4: Success probability as a function of the threshold.

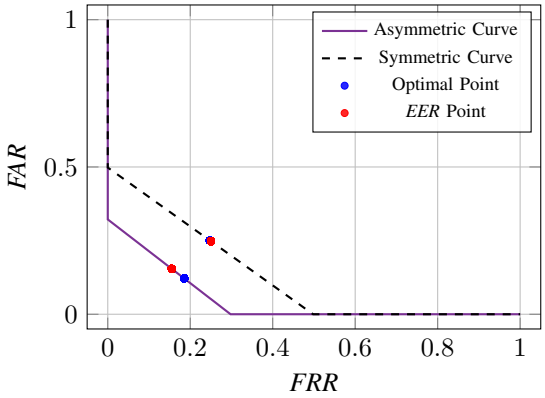


Figure 5: *FAR* vs. *FRR* operating curves for the asymmetric and symmetric distributions in (3) case, with optimal and *EER* points.

The maximum point represents the optimal threshold of the mechanism. From the optimal success probability expression, we can see that when an attacker distribution spans a wider interval, i.e., when $(a_2 - a_1)$ increases, the mechanism's success probability may significantly decrease. We calculate the *EER* point and show that it is not, in general, equal to the optimal point:

$$t_{EER} = \frac{a_1 \cdot u_1 - a_2 \cdot u_2}{a_1 - a_2 + u_1 - u_2}.$$

When the attacker's range lies withing the user's range, the optimal threshold lies in the boundry $t = a_2$ or inside the third interval $t = \frac{a_1+u_2}{2}$. These points are not equal to the *EER* and leads to different success probability. Similarly, in the overlap case, the optimal threshold lies in one of the boundries $t = u_1$, $t = a_2$ or inside the third interval where $t = \frac{a_1+u_2}{2}$. Thus, in both cases, the optimal threshold generally differs from the *EER*. However, when the user and attacker distributions are symmetric, i.e., when the intervals of each distribution are equal: $a_2 - a_1 = u_2 - u_1$, the *EER* expression coincides with the optimal threshold. Figure 5 illustrates this exception.

5.2. Quadratic Distributed Credential

Our goal is to model credentials whose distributions closely resemble those observed in real measurement data. To this end, we analyze a quadratic probability distribution function, which provides a second order approximation of the biometric credential that can also be seen in real fingerprint measurement data (§4). This choice is also consistent with the findings of Slobodan et al. [31] on Eigenfinger and Eigenpalm feature distributions. Accordingly, we define two probability density functions (PDFs). Let f_U denote the user distribution with roots α_1, α_2 and let f_A denote the attacker distribution with roots β_1, β_2. Denote by a, b, c and d, e, f the quadratic coefficients of the user and attacker distributions, respectively, hence:

$$f_U(s) = \begin{cases} a \cdot s^2 + b \cdot s + c & \alpha_1 \leq s \leq \alpha_2 \\ 0 & s > \alpha_2 \vee s < \alpha_1 \end{cases}$$

$$f_A(s) = \begin{cases} d \cdot s^2 + e \cdot s + f & \beta_1 \leq s \leq \beta_2 \\ 0 & s > \beta_2 \vee s < \beta_1 \end{cases}$$

To reduce the degrees of freedom of the quadratic distributions, we express the coefficients directly in terms of their roots. The integral of each PDF over its roots interval is set to 1, which enforces the normalization condition. This condition unsures that the resulting error metrics *FAR* and *FRR* are bounded to 1. Under this condition, the coefficients become fully determined. Accordingly, we define the coefficients as

$$\alpha_1 \cdot \alpha_2 = \frac{c}{a} \rightarrow a = \frac{c}{\alpha_1 \cdot \alpha_2}$$

$$\alpha_1 + \alpha_2 = -\frac{b}{a} \rightarrow b = -\frac{c \cdot (\alpha_1 + \alpha_2)}{\alpha_1 \cdot \alpha_2}$$

$$\beta_1 \cdot \beta_2 = \frac{f}{d} \rightarrow d = \frac{f}{\beta_1 \cdot \beta_2}$$

$$\beta_1 + \beta_2 = -\frac{e}{d} \rightarrow e = -\frac{f \cdot (\beta_1 + \beta_2)}{\beta_1 \cdot \beta_2}$$

$$\int_{\alpha_1}^{\alpha_2} (a \cdot x^2 + b \cdot x + c)dx = 1 \Rightarrow c = \frac{6 \cdot \alpha_1 \cdot \alpha_2}{(\alpha_1 - \alpha_2)^3}$$

$$\int_{\beta_1}^{\beta_2} (d \cdot x^2 + e \cdot x + f)dx = 1 \Rightarrow f = \frac{6 \cdot \beta_1 \cdot \beta_2}{(\beta_1 - \beta_2)^3}$$

Therefore, we can write the quadratic PDFs as

$$f_U(s) = \begin{cases} \dfrac{6(s - \alpha_1)(s - \alpha_2)}{(\alpha_1 - \alpha_2)^3} & \alpha_1 \leq s \leq \alpha_2 \\ 0, & \text{otherwise} \end{cases}$$

$$f_A(s) = \begin{cases} \dfrac{6(s - \beta_1)(s - \beta_2)}{(\beta_1 - \beta_2)^3} & \beta_1 \leq s \leq \beta_2 \\ 0, & \text{otherwise} \end{cases}$$

These distributions define the error metrics *FAR* and *FRR* as functions of the match threshold t. Denote

$$A(t) = \frac{2(\beta_2^3 - t^3) - 3(\beta_1 + \beta_2)(\beta_2^2 - t^2) + 6\beta_1\beta_2(\beta_2 - t)}{(\beta_1 - \beta_2)^3},$$

$$U(t) = \frac{2(t^3 - \alpha_1^3) - 3(\alpha_1 + \alpha_2)(t^2 - \alpha_1^2) + 6\alpha_1\alpha_2(t - \alpha_1)}{(\alpha_1 - \alpha_2)^3}.$$

$$\textit{FAR}(t) = \begin{cases} 1, & t \leq \beta_1, \\ A(t), & \beta_1 < t \leq \beta_2, \\ 0, & t > \beta_2, \end{cases}$$

and

$$\textit{FRR}(t) = \begin{cases} 0, & t \leq \alpha_1, \\ U(t), & \alpha_1 < t \leq \alpha_2, \\ 1, & t > \alpha_2. \end{cases}$$

Similarly to the analysis of the uniform distribution, the x axis represents the matching score s. Genuine users are expected to produce higher matching scores than attacker, thus we analyze three different cases:

(1) $\beta_2 < \alpha_1$ (Figure 6a): The two distributions are separated. Any threshold t chosen within the range $\beta_2 \leq t \leq \alpha_1$ yields zero *FAR* and *FRR*, and therefore achieves a mehcanism success probability of 1.

(2) $\alpha_1 < \beta_1 < \beta_2 < \alpha_2$ (Figure 6b): The attacker distribution bounds lies within the range of the user distribution bounds. The mechanisms probability of success reduces to the safe probability of the biometric credential, which is given by Equation 3 The *FAR* is one for any $t \leq \beta_1$ and zero for any $t \geq \beta_2$. Thus

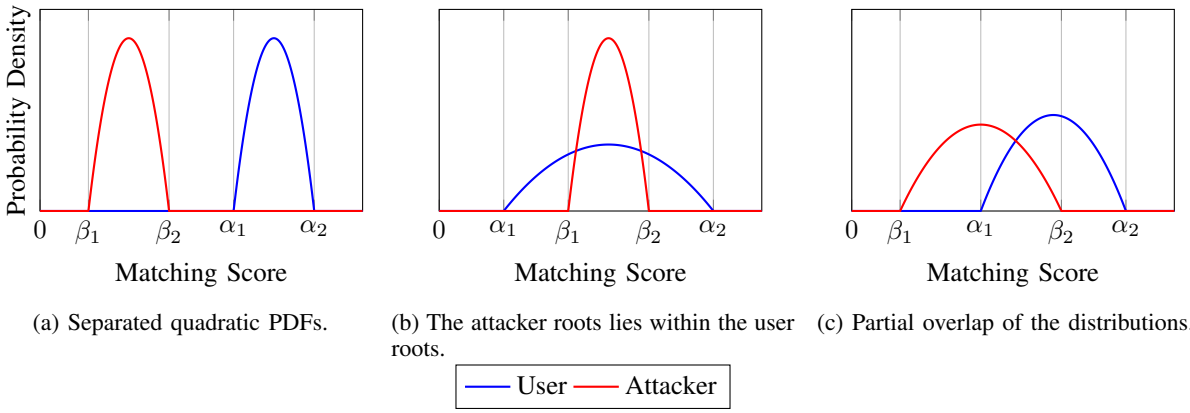
$$G(t) = -\frac{(t - \alpha_2)^2(t - \beta_1)^2}{(\alpha_1 - \alpha_2)^3(\beta_1 - \beta_2)^3} \cdot (2t - 3\alpha_1 + \alpha_2)(2t + \beta_1 - 3\beta_2).$$

$$P_{\text{success}}(t) = \begin{cases} 0, & t < \beta_1 \vee t > \alpha_2, \\ G(t), & \beta_1 \leq t \leq \beta_2, \\ -\dfrac{(t - \alpha_2)^2(2t - 3\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \leq t \leq \alpha_2. \end{cases}$$

(3) $\beta_1 \leq \alpha_1 \leq \beta_2 \leq \alpha_2$ (Figure 6c): The user and attacker distributions partially overlap. We mainatain same approach as the case (2) to calculate the probability of success. The *FAR* is one for any $t \leq \beta_1$ and zero for any $t \geq \beta_2$, and the *FRR* is zero for any $t \leq \alpha_1$ and one for any $t \geq \alpha_2$. Thus, in this case, we obtain 4 regions of the success probability function

$$P_{\text{success}}(t) = \begin{cases} 0, & t < \beta_1 \vee t > \alpha_2, \\ \dfrac{(t - \beta_1)^2(2t + \beta_1 - 3\beta_2)}{(\beta_1 - \beta_2)^3}, & \beta_1 \leq t \leq \alpha_1, \\ G(t), & \alpha_1 \leq t \leq \beta_2, \\ -\dfrac{(t - \alpha_2)^2(2t - 3\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \leq t \leq \alpha_2. \end{cases}$$

The second and third intervals in case (2) are identical to the third and fourth intervals in case (3) and therefore,



(a) Separated quadratic PDFs.

(b) The attacker roots lies within the user roots.

(c) Partial overlap of the distributions.

Figure 6: Illustration of three distinct cases of quadratic attacker and user score distributions. In case (1), the distributions are separated. Case (2) illustrates the case where the attacker distribution is fully contained within the user distribution, and case (3) illustrates partial overlap between the two distributions.

we analyze them together. We differentiate $P_{\text{success}}(t)$ and compute its critical points in every interval. By comparing the stationary values obtained in all valid regions, we identify the global maximum, which corresponds to the optimal threshold t_{opt}. Before writing the piecewise derivative, we define the polynomial coefficients and the following function:

$$
\begin{aligned}
&q = 4,\\
&m = -(5\alpha_1 + \alpha_2 + \beta_1 + 5\beta_2),\\
&n = \alpha_1(3\alpha_2 + \beta_1 + 6\beta_2) - \alpha_2^2 + \alpha_2\beta_2 - \beta_1^2 + 3\beta_1\beta_2,\\
&\ell = \alpha_1\big(\beta_1^2 - 3\beta_2(\alpha_2 + \beta_1)\big) + \alpha_2^2\beta_2,\\
&R(t) = -\frac{6(t-\alpha_2)(t-\beta_1)\big(qt^3 + mt^2 + nt + \ell\big)}{(\alpha_1-\alpha_2)^3(\beta_1-\beta_2)^3}.
\end{aligned}
$$

With these coefficients and function $R(t)$, the derivative of $P_{\text{success}}(t)$ is given by the following piecewise expression:

$$
\frac{dP_{\text{success}}}{dt} = \begin{cases} 0, & t < \beta_1 \ \vee \ t > \alpha_2, \\ \dfrac{6(t-\beta_1)(t-\beta_2)}{(\beta_1-\beta_2)^3}, & \beta_1 \le t \le \alpha_1, \\ R(t), & \alpha_1 \le t \le \beta_2, \\ -\dfrac{6(t-\alpha_1)(t-\alpha_2)}{(\alpha_1-\alpha_2)^3}, & \beta_2 \le t \le \alpha_2. \end{cases}
$$

In the second interval, the derivative has a single critical point at $t = \beta_1$. Since the derivative is strictly positive for all $t > \beta_1$, this point corresponds to a local minimum rather than a maximum. Similarly, in the fourth interval, the only critical point occurs at $t = \alpha_2$, and the derivative is strictly negative for all $t < \alpha_2$. Therefore, this interval also contains no maximum either. We therefore turn to the third interval. In this region, the derivative is the product of a cubic polynomial and two linear factors. The linear factors vanish at $t = \beta_1$ and $t = \alpha_2$, both of which lie outside the third interval. Consequently, the existence of critical points within this interval depends solely on the cubic term. Solving this cubic, reveals exactly one real root with the remaining two roots being complex. The closed form of the cubic solution cannot be simplified further more than a complex expression. Thus, we denote the solution as t_{cubic}.

By examining the sign of the derivative on both sides of the critical point t_{cubic}, we verify that it constitutes a global maximum of $P_{\text{safe}}(t)$. We denote this maximizing point by t_{opt}. Figure 8 illustrates this analysis.

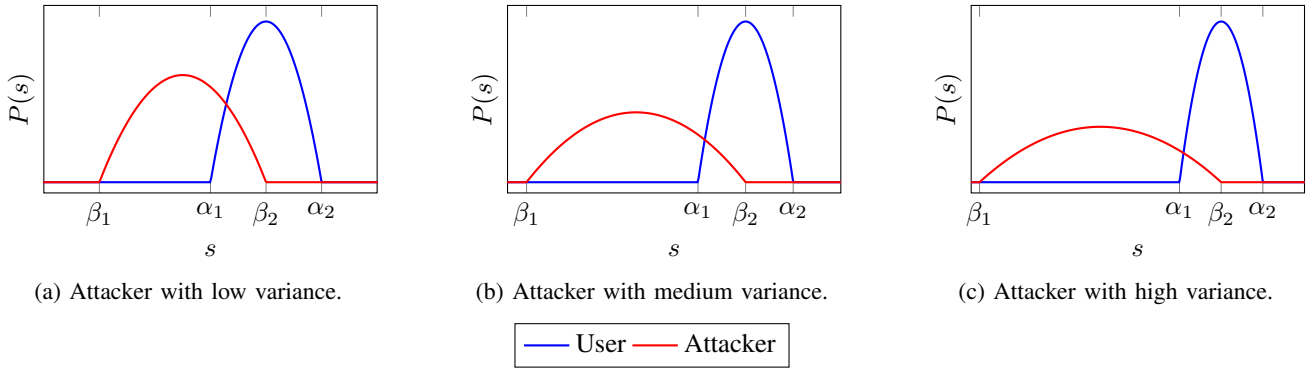
To demonstrate that this optimal threshold differs from the Equal Error Rate (*EER*) threshold, we computed the *EER* point separately as follows:

$$
t_{EER} = \frac{\alpha_1\beta_1 - \alpha_2\beta_2}{\alpha_1 - \alpha_2 + \beta_1 - \beta_2}
$$

We can observe that the *EER* threshold is not equal to the t_{opt}. To evaluate the difference between the t_{opt} and the *EER*, we illustrate different attacker distributions, each spanning a wider interval (β_1, β_2), which corresponds to increasing the variance, as shown in Figure 7. Comparing each case with its corresponding success probability curve, we observe that the optimal threshold t_{opt} becomes more distinct from the *EER* threshold t_{EER} as the variance of the attacker distribution increases and the more asymmetric the user and attacker distributions become.

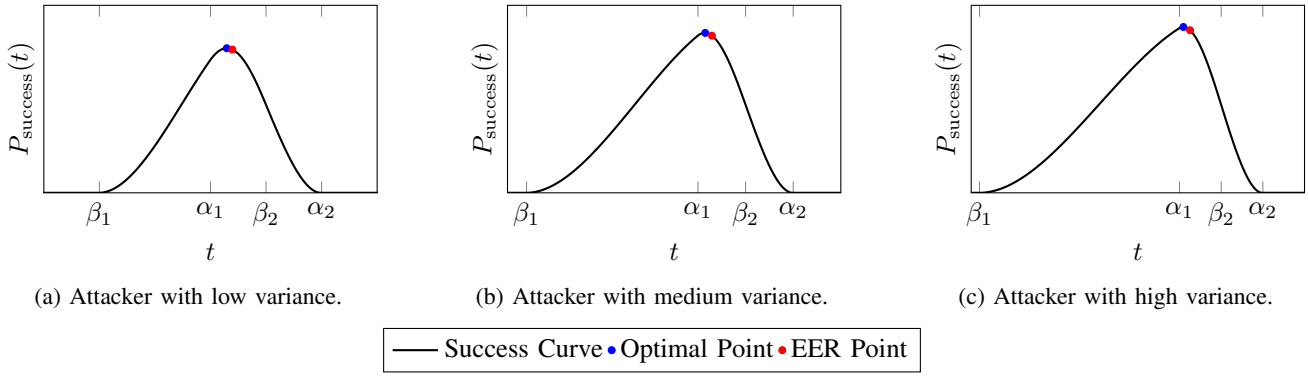
Moreover, we also observe that increasing the variance of an attacker distribution, shifts the optimal threshold t_{opt} towards α_1 reducing the *FRR* to zero, while the *EER* point remains fixed trying to balance between the *FAR* and *FRR*. This shift indicates that when increasing the variance, the optimal threshold prioritizes minimizing the false rejection rate, rather than balancing both error rates, as shown in Figure 9.

To quantify the effect of the variance on the deviation between the optimal threshold t_{opt} and *EER* threshold t_{EER}, we numerically computed this gap as a function of the attacker's variance. The simulation indicate that higher attacker variance enlarges the gap between the success probability at the optimal threshold and the success probability at the *EER* threshold as show in Figure 11. This finding highlights the potential success advantage gained by operating at t_{opt} rather



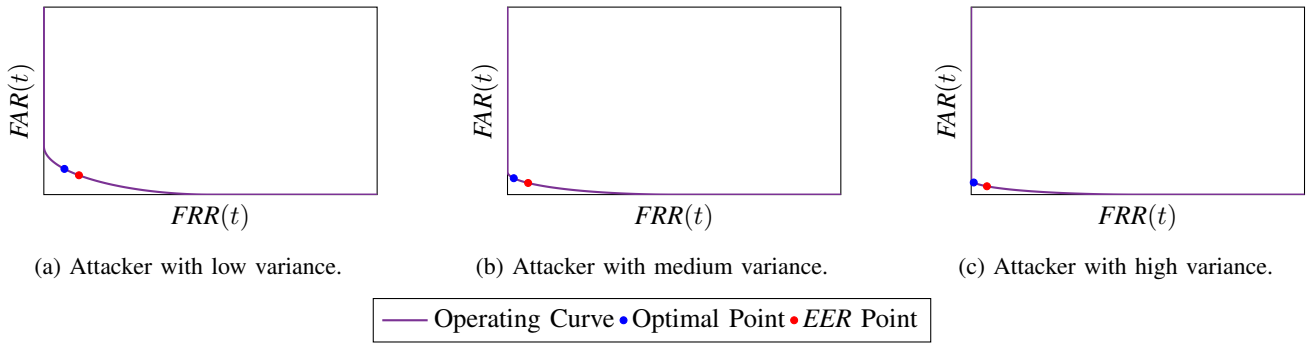
(a) Attacker with low variance. (b) Attacker with medium variance. (c) Attacker with high variance.

Figure 7: Quadratic user and attacker distributions.



(a) Attacker with low variance. (b) Attacker with medium variance. (c) Attacker with high variance.

Figure 8: Success probability for three quadratic attacker distributions.



(a) Attacker with low variance. (b) Attacker with medium variance. (c) Attacker with high variance.

Figure 9: *FAR* vs. *FRR* operating curves for different attacker distributions.

than at the *EER* point when an attacker distribution spans over wider intervals.

5.3. Single Credential Empirical Evaluation

We first evaluate a mechanism that uses a biometric credential only under the baseline attack, where attackers authenticate using their own fingerprints. Calculating the mechanism's success probability requires the biometric credential's fault probabilities, which depend on the *FAR* and *FRR*. We therefore follow the evaluation procedure used by Jain et al. [4], Cappelli et al. [6] and Slobodan et al. [31]. We produce histograms genuine and impostor comparisons, as shown in Figure 12aand calculate the *FAR* and *FRR* as functions of the threshold t, as shown in Figure 2.

Using these error rates, we calculate the biometric credential fault probabilities (Equation 3) and the mechanism's success probability (Equation 4). We find that the optimal threshold differs from the *EER* point and achieves a higher success probability, as shown in Figure 12b. However, the success probability of the mechanism, when operated at the optimal threshold, is not significantly greater than operating at the *EER* threshold.

We next consider a more challenging scenario in which attackers use forged fingerprints to impersonate legitimate users. Unlike the baseline attack, spoofing samples can

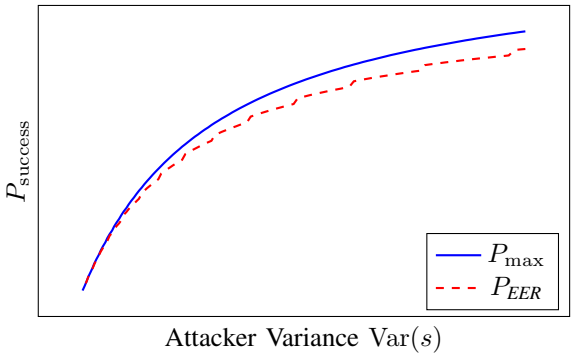


Figure 10: Success at the optimal and *EER* thresholds.

Figure 11: Comparison between the optimal threshold and the *EER* threshold as a function of the attacker's variance.

produce matching scores closer to those of genuine fingerprints, increasing the overlap between the distributions. In this evaluation, the *EER* increased significantly to 24.8%, indicating that the biometric credential is less effective at distinguishing users from attackers. We therefore evaluate the mechanism's success probability under this attack, as shown in Figure 13b. Nevertheless, despite the increased overlap between the user and the attacker distributions, the optimal and *EER* thresholds differ, but the improvement in success probability remains small. Under the baseline attack, the success probability increases from 92.42% to 92.54%, while under spoofing attacks, it increases from 50.64% to 50.88%. Thus, the EER heuristic performs well for a single biometric credential mechanism in our evaluation.

6. Two Credentials

We explore mechanisms that integrate two independently and identically distributed credentials. We numerically analyze two biometric credentials (§6.1) and symbolically analyze one biometric credential and one atomic credential (§6.2).

6.1. Two Biometric Credentials

To analyze the case of a mechanism M comprises two biometric credentials. We conduct a numerical study over several probability distribution functions. We start with a Gaussian distribution as a third-order approximation of a biometric credential's, as seen in Figure 2. To derive the optimal operating point for each possible mechanism composition, we maximize the respective success probabilities using a LBFGS optimization algorithm. We additionally compare the optimal operating point to the *EER* point, thereby characterizing the conditions under which the *AND* or *OR* mechanisms yields superior performance. We defined the user and attacker distributions as $f_U(s) = \mathcal{N}(\mu_u, \sigma_u^2)$ and $f_A(s) = \mathcal{N}(\mu_a, \sigma_a^2)$ respectively. We randomly generate the Gaussian parameters to construct a user and an attacker PDF's closely resemble the empirical findings of Slobodan [31]. We focus on the overlapping case between the user and attacker distributions, as this represents the common case among real-world scenarios. For the first credential as $\mu_u = 0.8$, $\sigma_u = 0.12$, $\mu_a = 0.3, \sigma_a = 1$ and for the second credential $\mu_u = 1.0$, $\sigma_u = 0.2$, $\mu_a = 0.4, \sigma_a = 0.18$ as shown in Figure 14.

For a mechanism M comprises two independently and identically distributed credentials, we defined mechanism's success under two scenarios: a. *AND*: The user can access both credentials, while the attacker cannot access least one of the credentials. The mechanism succeeds when both credentials are in the safe state or when one is in the safe state and the other is in the leak state. Conversely, if either credential is in the lost or stolen state, the user cannot satisfy the mechanism, and the mechanism fails.

$$P_{\text{Success}}(M^{\text{AND}}) = \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{safe}} + \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{leak}} + \text{P}^2_{\text{safe}} \cdot \text{P}^1_{\text{leak}} \tag{5}$$

b. *OR*: The user can access at least one credential, while the attacker cannot access either of them. The mechanism succeeds when both credentials are in the safe state or when one is in the safe state and the other is in the lost state. If either credential is leak or stolen, an attacker can satisfy the mechanism and thus the mechanism fails.

$$P_{\text{Success}}(M^{\text{OR}}) = \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{safe}} + \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{loss}} + \text{P}^2_{\text{safe}} \cdot \text{P}^1_{\text{loss}} \tag{6}$$

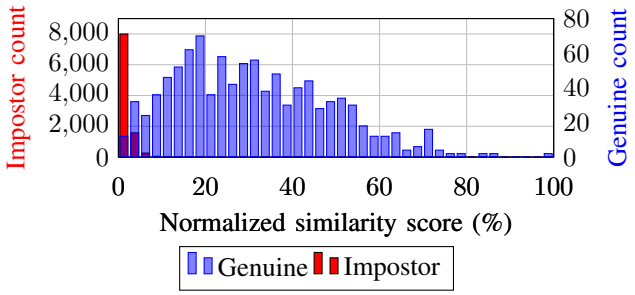
We calculate the error metrics of each credential, which are given by Equations 1 and 2 as a function of the threshold t. We then use these metrics to derive the credential's fault probabilities, as defined in Section 3.2.

$$\mathit{FAR}(t) = \int_{-\infty}^{t} \frac{1}{\sigma_{a,i}\sqrt{2\pi}} \exp\left(-\frac{(s-\mu_{a,i})^2}{2\sigma_{a,i}^2}\right) ds, \text{ and}$$

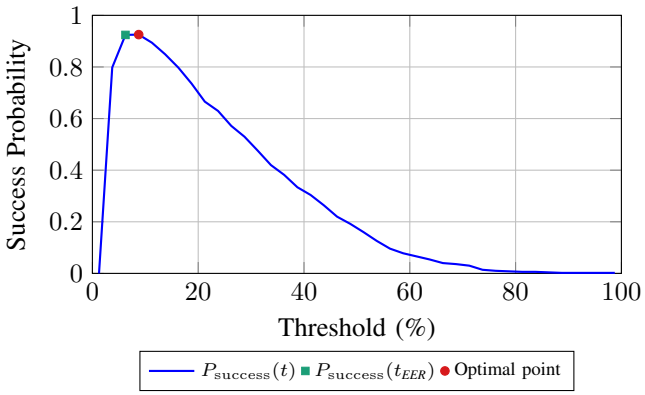
$$\mathit{FRR}(t) = \int_{t}^{\infty} \frac{1}{\sigma_{u,i}\sqrt{2\pi}} \exp\left(-\frac{(s-\mu_{u,i})^2}{2\sigma_{u,i}^2}\right) ds.$$

We analyze both *AND* and *OR* mechanisms success probability given by Equations 5 and 6 and find their optimal threshold t, using LBFGS optimization algorithm. We also compute the *EER* threshold of each credential and find the corresponding success probability at this point.

The optimal threshold for both mechaninsms' types differ from the *EER*, yielding higher success probabilities than those obtained at the *EER*. Moreover, we find that the optimal points of both the *AND* and *OR* mechanisms are not necessarily the same as the optimal threshold if used individually. This finding suggest that integrators of a biometric credential should determine its threshold as function of the other credentials integrated into the mechanism. Figure 15 illustrates the *FAR* vs. *FRR* curves for both credentials, and marks the optimal points of both *AND* and *OR* mechanisms, as well as the *EER* and the standalone optimum of the biometric credential. The figure highlights the gap between different threhsolds. In the first credential, the threshold leads to high *FRR* and low *FAR* values, while in the second credential the threshold leads to low *FRR* and high *FAR*

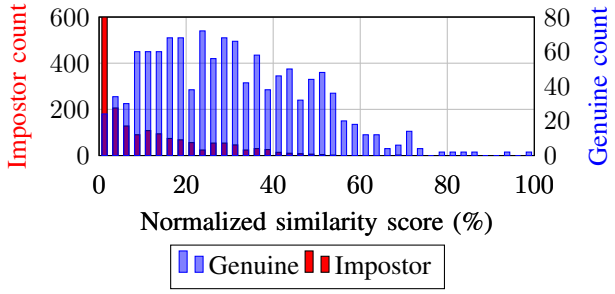


(a) Score histogram for genuine and impostor comparisons.

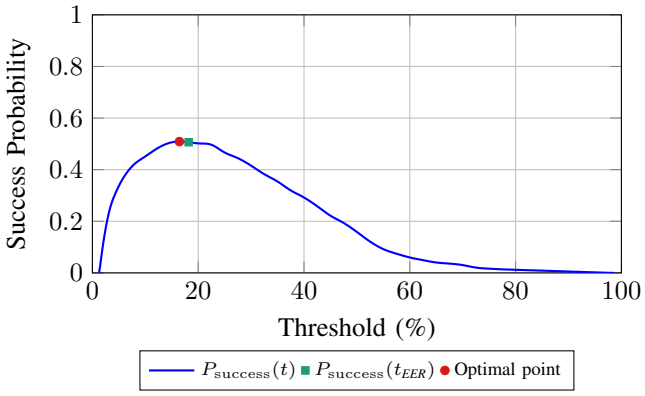


(b) Single biometric credential mechanism's success.

Figure 12: Baseline attack evaluation of a mechanism that uses a biometric credential only.



(a) Score histogram for genuine and impostor comparisons.



(b) Single biometric credential mechanism's success.

Figure 13: Spoofing attack evaluation of a mechanism that uses a biometric credential only.

values. This demonstrates that the *AND* and *OR* mechanisms balance the two error rates differently across credentials.

We also find that the integration of a biometric credential to a mechanism already comprises a biometric credential, improves both *AND* and *OR* mechanism success. Hence, integrating two biometric credentials into a mechanism improves its success over using them individually. Figure 16 illustrate the success probability of both *AND* and *OR* mechanisms. We fix a credential to its optimum and show how the success probability varies as a function of the thresholds of the other credential.

We numerically analyze first and second order approximations of the biometric credential's distribution and calculate the success probabilities of both *AND* and *OR* mechanisms at different thresholds. We quantify the improvement achieved by integrating two biometric credentials and the improvement of using the optimal threshold over the *EER* threshold. In conclusion, all numerical results are summarized in Table 1.

From Table 1, we observe that the optimal operating point yields a higher success probability than the *EER* point by between 5% to 7%.

6.2. One Atomic and One Biometric Credential

We start by analyzing the case of an *AND* mechanism comprises an atomic credential and a biometric credential. We define the atomic credential with a constant value of each probabilities' states as: P^1_{safe}, P^1_{loss}, P^1_{leak}, P^1_{theft} corrsponding respectively to each one of the states: safe, loss, leak and theft. We define the biometric credential by its PDFs for the user and the attacker, f_U and f_A respectively. We consider uniform distribution as a first-order approximation of a biometric credential's distribution. Denote by u_1, u_2 the user's PDF bounds and a_1, a_2 the attacker's PDF bounds, as shown in Figure 3. The

Distribution	$P_{c_1}(\text{safe})$	$P_{c_2}(\text{safe})$	$P_{\text{AND}}(t_{EER})$	$P_{\text{OR}}(t_{EER})$	$P_{\text{AND}}(t_1^{\text{safe}})$	$P_{\text{OR}}(t_2^{\text{safe}})$	$\mathbf{P}_{\text{AND}}(\mathbf{t}_{\text{opt}})$	$\mathbf{P}_{\text{OR}}(\mathbf{t}_{\text{opt}})$	Improvement
Gaussian	0.937	0.889	0.911	0.911	0.912	0.911	**0.964**	**0.963**	5.90% / 5.66%
Uniform	1.000	0.916	0.956	0.956	0.917	0.999	**1.000**	**1.000**	4.63% / 4.61%
Parabolic	0.959	0.887	0.921	0.921	0.920	0.923	**0.987**	**0.985**	7.25% / 6.99%

TABLE 1: Mechanism success under two biometric credentials with Gaussian, Uniform and Parabolic distributions at four reference operating points: safe success of a single biometric credential mechanism, *EER*-pair (t_1^{EER}, t_2^{EER}), individual optimum pair $(t_1^{\text{safe}}, t_2^{\text{safe}})$, and joint optima (AND at $(t_1^*, t_2^*)_{\text{AND}}$ and OR at $(t_1^*, t_2^*)_{\text{OR}}$) . The improvement is measured between the optimal operating point and the *EER* point.

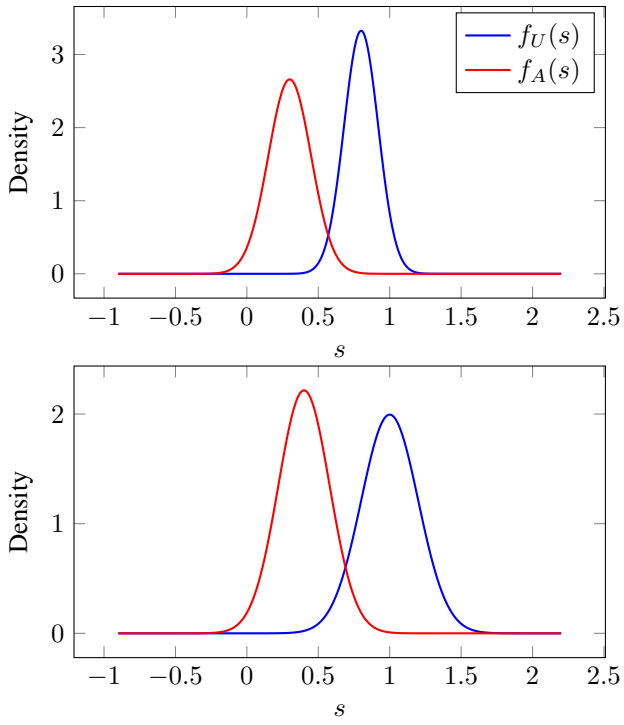


Figure 14: Gaussian user and attacker distributions.

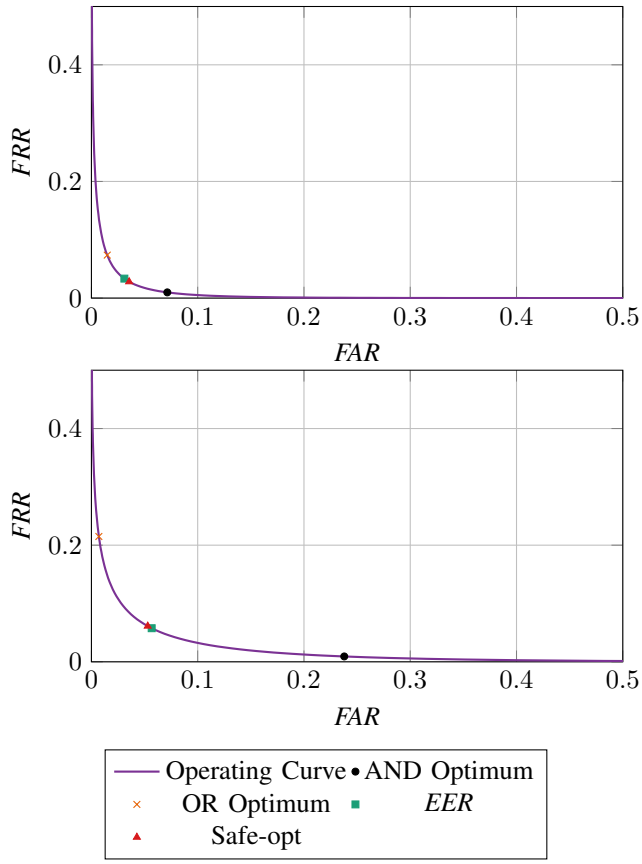


Figure 15: *FAR* vs. *FRR* operating curves for two biometric credentials.

state probabilities of the biometric credential are defined by the error metrics *FAR* and *FRR* as done in Section 3.2. Moreover, we consider the overlapping case between the user and attacker distributions, as this represents the common case among real-world scenarios. Thus, we assume that $a_1 < u_1 < a_2 < u_2$. We set the state probabilities of an atomic and a biometric credential into Equation 5. Thus, the probability of success is a function of the threshold t and divided into 5 intervals. Denote:

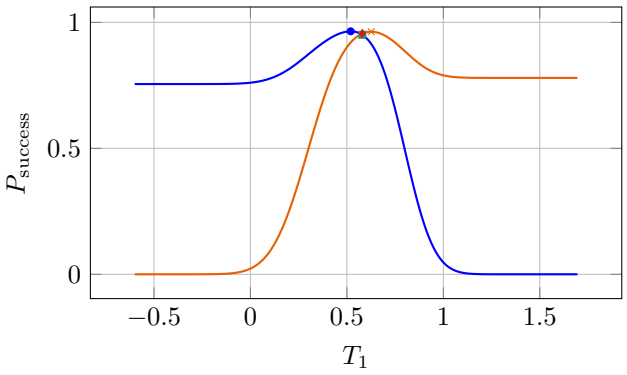
$$L(t) = \frac{u_2 - t}{u_2 - u_1}\left[P^1_{\text{safe}} + P^1_{\text{leak}}(1 - \frac{a_2 - t}{a_2 - a_1})\right].$$

$$P_{\text{success}}(t) = \begin{cases} P^1_{\text{safe}}, & t \le a_1 \\ P^1_{\text{safe}} + P^1_{\text{leak}}(1 - \dfrac{a_2 - t}{a_2 - a_1}), & a_1 < t \le u_1 \\ L(t), & u_1 < t \le a_2 \\ \dfrac{u_2 - t}{u_2 - u_1} \cdot (P^1_{\text{safe}} + P^1_{\text{leak}}), & a_2 < t \le u_2 \\ 0, & t > u_2. \end{cases}$$

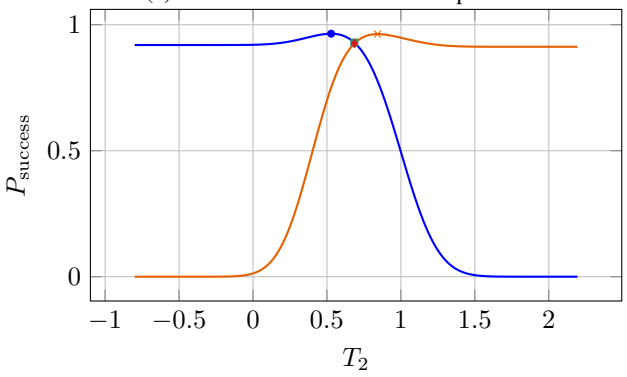
We analyze the success probability $P_{\text{success}}(t)$ by partitioning the threshold domain into five intervals.

The first interval, $t \le a_1$, corresponds to the case in which the threshold is lower than the minimum value of the attacker distribution. In this regime, the *FAR* is equal to 1, while the *FRR* is equal to 0. Consequently, the safe probability of the biometric credential is zero, and the success probability converges to P^1_{safe}. Since P^1_{safe} is constant in this interval, no interior maximum exists.

The second interval, $a_1 < t \le u_1$, corresponds to the case in which the threshold lies between the minimum value of the attacker distribution and the minimum value of the user distribution. In this region, the *FAR* decreases linearly as a function of t, while the *FRR* remains equal to 0. As a result, P^2_{safe} and P^2_{leak} of the biometric credential are $1 -$



(a) Biometric credential 1 sweep.



(b) Biometric credential 2 sweep.

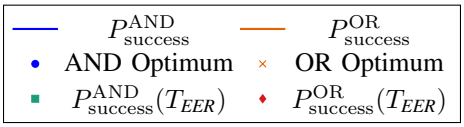


Figure 16: Success probability as a function of the threshold for two biometric credentials.

FAR(t) and *FAR*(t), respectively. In this interval, $P_{\text{success}}(t)$ is a strictly increasing function, as its derivative is a constant positive value, and therefore the maximum is attained at the boundary point: $t_{\text{opt},2} = u_1$.

The third interval, $u_1 < t \leq a_2$, corresponds to the case in which the threshold lies between the minimum value of the user distribution and the maximum value of the attacker distribution. In this region, both *FAR* and *FRR* vary linearly with t, resulting in a success probability that is quadratic as a function of t. Differentiating $P_{\text{success}}(t)$ yields a single critical point inside the interval,

$$t'_{\text{opt}} = \frac{u_2 + a_2 - (a_2 - a_1)\,\dfrac{P^1_{\text{safe}} + P^1_{\text{leak}}}{P^1_{\text{leak}}}}{2}$$

Since this expression is valid only for the interval $u_1 < t \leq a_2$, the effective maximizer within this interval is obtained by clipping the optimum to the interval boundaries, as follows:

$$t_{\text{opt},3} = \min\big(a_2,\ \max\big(u_1,\ t'_{\text{opt}}\big)\big)\,.$$

In particular, if $t_{\text{opt},3} < u_1$, the maximum over this interval is attained at $t = u_1$, whereas if $t_{\text{opt},3} > a_2$, the maximum is attained at $t = a_2$. Evaluating the success probability at the user threshold $t = u_1$ yields

$$P_{\text{success}}(u_1) = P^1_{\text{safe}} + P^1_{\text{leak}}(1 - \frac{a_2 - u_1}{a_2 - a_1}).$$

Since we are analyzing the overlapping case where $u_1 > a_1$, it follows that

$$P_{\text{success}}(u_1) > P^1_{\text{safe}},$$

Demonstrating that the integration of the biometric credential strictly improves the success of the mechanism compared to using the atomic credential alone.

By examining the optimal point in this interval under *AND* mechanism, we can deduce that as the variance of the attacker distribution increases, the optimal threshold $t_{\text{opt},3}$ shifts toward the lower boundry of the interval, u_1. Moreover, for an atomic credential with high safe values P^1_{safe}, and low leak values P^1_{leak}, the optimal threshold shifts toward the lower boundary as well. Thus, in such case, the success of the mechnaims becomes mainly dependent on the atomic credential. Conversely, an atomic credential with high leak values P^1_{leak} and low safe values P^1_{safe} causes the optimal threshold to shift toward the upper boundary of the interval, a_2. In this case, the success of the mechanism relies more heavily on the biometric credential.

The fourth interval, $a_2 < t \leq u_2$, corresponds to the case in which the threshold exceeds the maximum value of the attacker distribution. In this regime, the *FAR* is equal to 0, while the *FRR* increases linearly with t. Consequently, $P_{\text{success}}(t)$ is a strictly decreasing function in this interval, and the maximum is attained at the boundary point

$$t_{\text{opt},4} = a_2.$$

Finally, the fifth interval, $t > u_2$, corresponds to the case in which the threshold exceeds the maximum value of the user and the attacker distributions, leaving the success probability equal to 0.

To determine the global optimal operating point, we compare all stationary points and boundary maxima obtained in the valid intervals. The optimal threshold is achieved in the third interval, where the attacker and user distributions overlap and the success function attains an interior maximum. Substituting $t_{\text{opt}} = t_{\text{opt},3}$ into $P_{\text{success}}(t)$ yields

$$P_{\text{success}}(t_{\text{opt},3}) = \frac{\big(a_2 P^1_{\text{safe}} - a_1\big(P^1_{\text{safe}} + P^1_{\text{leak}}\big) + P^1_{\text{leak}}\,u_2\big)^2}{4P^1_{\text{leak}}(a_2 - a_1)(u_2 - u_1)}.$$

We do similar analysis for the *OR* mechanism. We find the probability of success for the *OR* mechanism. Denote:

$$Q(t) = \frac{t - a_1}{a_2 - a_1}\left[P^1_{\text{safe}} + P^1_{\text{loss}}(1 - \frac{t - u_1}{u_2 - u_1})\right].$$

$$P_{\text{success}}(t) = \begin{cases} 0, & t \le a_1 \\ \dfrac{t-a_1}{a_2-a_1}\,(P^1_{\text{safe}} + P^1_{\text{loss}}), & a_1 < t \le u_1 \\ Q(t), & u_1 < t \le a_2 \\ P^1_{\text{safe}} + P^1_{\text{loss}}(1 - \dfrac{t-u_1}{u_2-u_1}), & a_2 < t \le u_2 \\ P^1_{\text{safe}}, & t > u_2. \end{cases}$$

We illustrate the mechanism's success probability for both *OR* mechanism in figure 17a and *AND* mechanism in Figure 17b. The figure marks the optimal operating point and the *EER* point on each curve demonstrating significant improvement from *EER* using optimal threshold. Moreover, the figure shows that when integrating a biometric credential and determining the threshold t to *EER*, the integration reduces mechanism's success compared to using the atomic credential alone. However, by optimizing the threshold, the integration of the biometric credential improves the mechanism's success over using the atomic credential alone. By differentiating every interval and identifying the critical points, we found that the optimal threshold that leads to the maximum probability of success lies in the third interval where $u_1 < t \le a_2$:

$$t_{\text{opt}} = \frac{a_1 \cdot P^1_{\text{loss}} + u_2 \cdot (P^1_{\text{safe}} + P^1_{\text{loss}}) - P^1_{\text{safe}} \cdot u_1}{2P^1_{\text{loss}}}$$

And the maximum probability of success at this point is:

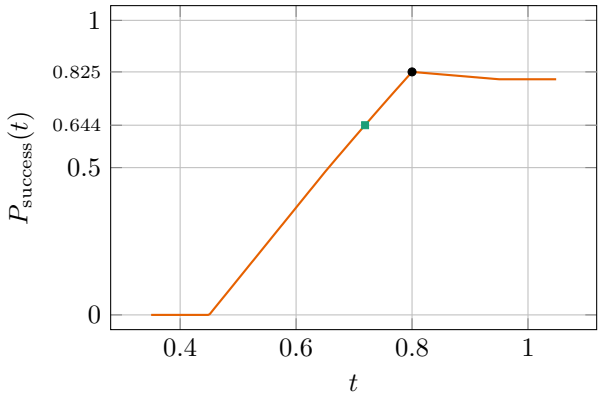
$$P_{\text{success}}(t_{\text{opt}}) = \frac{\left(a_1 P^1_{\text{loss}} - u_2(P^1_{\text{safe}} + P^1_{\text{loss}}) + P^1_{\text{safe}} u_1\right)^2}{4P^1_{\text{loss}}(a_2 - a_1)(u_2 - u_1)}$$

we calculate the *EER* point and show that the optimal operating point t_{opt} differs from the equal error rate (*EER*) point t_{EER}.

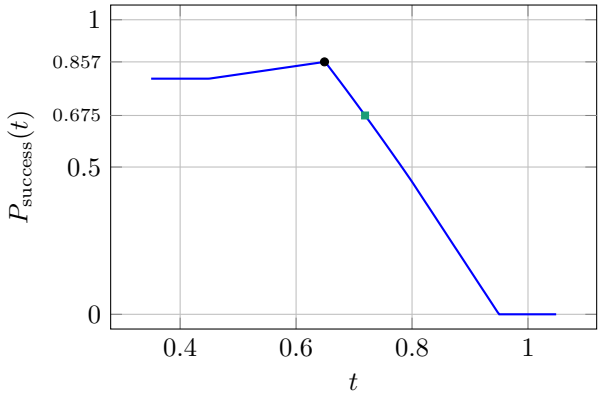
$$t_{EER} = \frac{a_1 u_1 - a_2 u_2}{a_1 - a_2 + u_1 - u_2} \neq t_{\text{opt}}.$$

6.3. Two Credentials Empirical Evaluation

We use the same biometric credential evaluated in (§5.3), and first consider the baseline attack scenario, where attackers use their own fingerprints to authenticate. The mechanisms combine this biometric credential with an atomic credential whose fault probabilities are arbitrarily set to $P^1_{\text{safe}} = 0.75$, $P^1_{\text{leak}} = 0.1$, $P^1_{\text{loss}} = 0.1$, and $P^1_{\text{stolen}} = 0.05$. We then calculate the success probabilities of the *AND mechanism* (Equation 5), and the *OR mechanism* (Equation 6). We illustrate these probabilities as functions of the threshold t in Figure 18. We show that the optimal threshold for each mechanism differs from the *EER* threshold. Moreover, both mechanisms operating at their optimal thresholds achieve success probabilities that are 8-9% higher than that of a mechanism using only the atomic credential, demonstrating the benefit of combining the two credentials.



(a) OR mechanism.



(b) AND mechanism.



Figure 17: Success probability as a function of the threshold for the OR and AND mechanisms composed of an atomic credential with state probabilities $P^1_{\text{safe}} = 0.8$, $P^1_{\text{loss}} = 0.05$, $P^1_{\text{leak}} = 0.1$, $P^1_{\text{theft}} = 0.05$, and a biometric credential.

However, in this attack scenario, optimizing the thresholds provides only a small improvement over the EER threshold for both mechanisms, since the biometric credential already has a very low EER, indicating limited overlap between the genuine and impostor distributions.

We next evaluate the two mechanisms under spoofing attacks, where attackers use forged fingerprints to impersonate legitimate users. As shown in the previous subsection, spoofing increases the overlap between the genuine and impostor score distributions, making authentication more challenging. We calculate the success probabilities of both mechanisms under this attack, as shown in Figure 13b. We find that the optimal threshold reduces the failure probability between 20%-72% compared to the *EER* threshold for the two mechanisms. Moreover, a mechanism combining both credentials at the *EER* threshold may achieve a lower success probability than a mechanism using either credential alone. These results demonstrate that using a non-optimal biometric threshold can reduce the success probability when combining credentials in an authentication mechanism. The

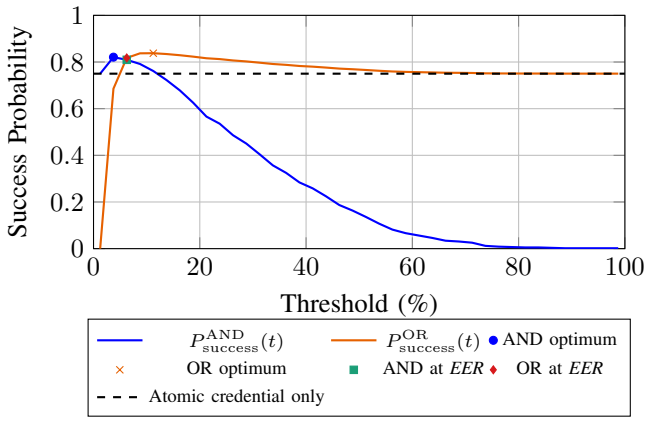


Figure 18: Baseline attack. A biometric credential integrated with an atomic credential into AND and OR mechanisms. The AND and OR mechanisms achieve success probabilities of 0.78 and 0.82 at the *EER* point, and 0.81 and 0.84 at their respective optimal operating points.

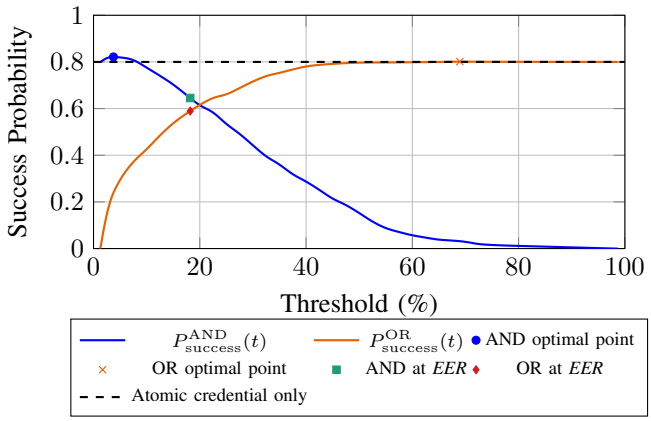


Figure 19: Spoofing attack. A biometric credential integrated with an atomic credential into AND and OR mechanisms. The AND mechanism achieves a maximal success probability of 0.834, while the OR mechanism achieves 0.816. At the *EER* operating point, the success probabilities are 0.713 and 0.649 for the AND and OR mechanisms, respectively.

results are summarized in Table 2.

7. Characterizing Success Improvement

We characterize the conditions under which a mechanism that uses both a biometric credential and an atomic credential mechanism achieves greater success than a mechanism that uses only the atomic credential. We show that any mechanism using a single atomic credential with $P^1_{\text{loss}} = P^1_{\text{leak}} = 0$ and a biometric credential does not achieve greater success probability than a mechanism using only this atomic credential (§7.1). Next, we present a necessary and sufficient condition for such improvement (§7.2), extend the results to multiple atomic credentials, and derive a more intuitive sufficient condition.

7.1. No Improvement In Mechanism Success

We first consider an atomic credential that is always either safe or stolen, never lost or leaked. This case admits no benefit from producing a mechanism that uses an atomic credential in this setting and a biometric credential. The following proposition formalizes this result.

Proposition 1. *Let c_1 be an atomic credential with $P^1_{loss} = P^1_{leak} = 0$, c_2 be a biometric credential with user distribution $f_U(t)$ and attacker distribution $f_A(t)$, and M be a mechanism using c_1. Then, for a mechanism $\bar{M}$ using c_1, c_2 in* AND *or* OR *compositions it holds that*

$$P^{\bar{M}}_{\text{success}} \leq P^{M}_{\text{success}}.$$

Proof. The mechanism $\bar{M}$ uses one atomic credential and one biometric credential under two possible compositions (see §6). An *AND mechanism* has success probability described by Equation 5, and an *OR mechanism* has success probability described by Equation 6. For both equations, $P^2_{\text{safe}}(t)$ denotes the safe probability of the biometric credential as a function of the operating threshold t. Since the state probabilities of the atomic credential satisfy $P^1_{\text{leak}} = P^1_{\text{loss}} = 0$, the success probability of $\bar{M}$ reduces in both cases to

$$P_{\text{success}}(t) = P^1_{\text{safe}} \cdot P^2_{\text{safe}}(t).$$

Each state probability of the biometric credential is strictly less than one; in particular, $P^2_{\text{safe}}(t) \leq 1$ for all values of t. Therefore, $P_{\text{success}}(t) \leq P^1_{\text{safe}}$ and thus, adding a biometric credential does not produce a new mechanism $\bar{M}$ whose success probability is greater than that of a single atomic credential mechanism. □

7.2. Necessary and Sufficient Condition for Integrating Biometric Credentials

We turn to the general case of an atomic credential that can also be lost or leaked. Then, we identify a biometric credential that is *asymptotically separable*. We show that this is a necessary and sufficient condition for a mechanism using an atomic credential and a biometric credential to achieve strictly greater success probability than a mechanism using only the atomic credential. Intuitively, asymptotic separability means that there exists a threshold t such that the error rate is negligible with respect to the correctness rate, making the user and attacker distributions effectively separated.

Definition 6. *A user distribution function $f_U(s)$ and an attacker distribution function $f_A(s)$ are* asymptotically separable *if*

$$\inf_t \min\left\{ \frac{FAR(t)}{1 - FRR(t)}, \frac{FRR(t)}{1 - FAR(t)} \right\} = 0.$$

We now prove that asymptotic separability of a biometric credential is a sufficient condition.

TABLE 2: Comparison between the success probabilities obtained at the EER threshold and the maximal success probabilities. Bold EER values indicate the worse composition, whereas bold maximal values indicate the best composition. The bold safe column indicating the success probability of a mechanism uses an atomic credential only. The improvement column shows the percentage of improvement between the failure probability at the optimal mechanism success and the failure probability at the maximal mechanism success at EER threshold.

$\mathbf{P}_{\text{safe}}$	P_{loss}	P_{leak}	P_{theft}	AND at EER	OR at EER	Max AND	Max OR	Improvement
0.60	0.30	0.05	0.05	**0.45**	0.58	0.60	**0.66**	20.25%
0.60	0.05	0.30	0.05	0.57	**0.45**	**0.68**	0.60	24.94%
0.75	0.15	0.05	0.05	**0.56**	0.61	0.75	**0.77**	40.39%
0.82	0.05	0.08	0.05	0.62	**0.61**	**0.824**	0.82	53.20%
0.90	0.00	0.05	0.05	0.66	**0.64**	0.89	**0.905**	72.19%

Proposition 2. *Let c_1 be an atomic credential with $P^1_{safe} > 0$ and at least one of $P^1_{leak}, P^1_{loss} > 0$, c_2 be a biometric credential whose user distribution $f_U(t)$ and attacker distribution $f_A(t)$ are asymptotically separable and M be a mechanism using c_1. Then, there exists a mechanism $\bar{M}$ which is an* AND *or* OR *composition using c_1 and c_2, such that*

$$P^{\bar{M}}_{\text{success}} > P^{M}_{\text{success}}$$

Proof. The success probability of a mechanism comprising an atomic credential only is P^1_{safe} (Equation 4). A biometric credential is characterized by four state probabilities, which are functions of t: $P^2_{\text{safe}}(t)$, $P^2_{\text{loss}}(t)$, $P^2_{\text{leak}}(t)$, $P^2_{\text{theft}}(t)$. As previously presented in the model (§3.2), these probabilities are functions of *FAR* and *FRR*.

$$\begin{aligned}
P^2_{\text{safe}}(t) &= 1 - \mathit{FAR}(t) - \mathit{FRR}(t) + \mathit{FAR}(t) \cdot \mathit{FRR}(t),\\
P^2_{\text{loss}}(t) &= \mathit{FRR}(t)\big(1 - \mathit{FAR}(t)\big),\\
P^2_{\text{leak}}(t) &= \mathit{FAR}(t)\big(1 - \mathit{FRR}(t)\big),\\
P^2_{\text{theft}}(t) &= \mathit{FAR}(t) \cdot \mathit{FRR}(t).
\end{aligned}$$

A mechanism $\bar{M}$ using c_1 and c_2 has two possible compositions: an *AND mechanism* with success probability described by Equation 5, or an *OR mechanism* with success probability described by Equation 6. Thus, by placing the biometric credential in each configuration, we express the success probability of $\bar{M}$ as a function of the operating threshold t:

$$P^{\text{AND}}_{\text{success}}(t) = \big(1 - \mathit{FRR}(t)\big)\Big(P^1_{\text{safe}} + P^1_{\text{leak}}\big(1 - \mathit{FAR}(t)\big)\Big), \tag{7}$$

$$P^{\text{OR}}_{\text{success}}(t) = \big(1 - \mathit{FAR}(t)\big)\Big(P^1_{\text{safe}} + P^1_{\text{loss}}\big(1 - \mathit{FRR}(t)\big)\Big). \tag{8}$$

We need to prove that there exists an operating threshold t such that $P^{\text{AND}}_{\text{success}}(t) > P^1_{\text{safe}}$ or $P^{\text{OR}}_{\text{success}}(t) > P^1_{\text{safe}}$. We plug Equation 7 into $P^{\text{AND}}_{\text{success}}(t) > P^1_{\text{safe}}$ to achieve a sufficient condition for this inequality. Thus:

$$\begin{aligned}
&P^1_{\text{safe}} < (1 - \mathit{FRR}(t))\Big(P^1_{\text{safe}} + P^1_{\text{leak}}(1 - \mathit{FAR}(t))\Big) \iff\\
&\frac{P^1_{\text{safe}}}{P^1_{\text{safe}} + P^1_{\text{leak}} \cdot (1 - \mathit{FAR}(t))} < 1 - \mathit{FRR}(t) \iff\\
&\mathit{FRR}(t) < 1 - \frac{P^1_{\text{safe}}}{P^1_{\text{safe}} + P^1_{\text{leak}} \cdot (1 - \mathit{FAR}(t))} \iff\\
&\mathit{FRR}(t) < \frac{P^1_{\text{leak}} \cdot (1 - \mathit{FAR}(t))}{P^1_{\text{safe}} + P^1_{\text{leak}} \cdot (1 - \mathit{FAR}(t))} \iff\\
&\frac{\mathit{FRR}(t)}{1 - \mathit{FAR}(t)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}} \cdot (1 - \mathit{FAR}(t))}.
\end{aligned} \tag{9}$$

Division by $(1-\mathit{FAR}(t))$ in the fourth step is valid under the imposed constraint that $P^2_{safe} > 0$, which implies $\mathit{FAR} < 1$ and $\mathit{FRR} < 1$. Since $P^1_{\text{leak}} \cdot (1-\mathit{FAR}(t)) \leq P^1_{\text{leak}}$, a sufficient condition for the above inequality is

$$\frac{\mathit{FRR}(t)}{1 - \mathit{FAR}(t)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}}} \tag{10}$$

Similarly, we plug Equation 8 into $P^{\text{OR}}_{\text{success}}(t) > P^1_{\text{safe}}$, and deduce that this inequality holds when:

$$\frac{\mathit{FAR}(t)}{1 - \mathit{FRR}(t)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}} \cdot (1 - \mathit{FRR}(t))} \tag{11}$$

Division by $(1 - \mathit{FRR}(t))$ is valid under the same imposed constraint that $P^2_{safe} > 0$. Since $P^1_{\text{loss}} \cdot (1 - \mathit{FRR}(t)) \leq P^1_{\text{loss}}$, a sufficient condition for the above inequality is

$$\frac{\mathit{FAR}(t)}{1 - \mathit{FRR}(t)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}}} \tag{12}$$

By the definition of asymptotic separability, there exists an operating threshold t such that

$$\inf_t \min\left\{\frac{\mathit{FAR}(t)}{1 - \mathit{FRR}(t)}, \frac{\mathit{FRR}(t)}{1 - \mathit{FAR}(t)}\right\} = 0.$$

Equivalently, for all $\varepsilon > 0$, there exists t such that either $\frac{\mathit{FAR}(t)}{1-\mathit{FRR}(t)} < \varepsilon$ or $\frac{\mathit{FRR}(t)}{1-\mathit{FAR}(t)} < \varepsilon$. Thus, we can choose ε that equals the right hand side of the sufficient condition where the AND mechanism is greater than M (Equation 10) or the the right hand side where the OR mechanism is greater than M (Equation 12). Hence, by definition of

asymptotic separability we know that a t that satisfies the inequality exists. More precisely, if

$$\inf_t \frac{\mathit{FAR}(t)}{1-\mathit{FRR}(t)} = 0,$$

then the OR mechanism composition is preferable, whereas if

$$\inf_t \frac{\mathit{FRR}(t)}{1-\mathit{FAR}(t)} = 0,$$

then the AND mechanism composition is preferable. Therefore, a mechanism $\bar{M}$ uses an asymptotically separable biometric credential and an atomic credential achieves a greater success probability than that of M. □

We showed that *asymptotic separability* is a sufficient condition. We now show it is a necessary condition for producing a new mechanism $\bar{M}$ with greater success probability than that of M.

Proposition 3. *Let c_2 be a biometric credential with user distribution $f_U(t)$ and attacker distribution $f_A(t)$. If for every atomic credential c_1 with $P^1_{safe} > 0$ and at least one of $P^1_{leak}, P^1_{loss} > 0$, there exists a threshold t and a mechanism $\bar{M}$ formed as an* AND *or* OR *composition such that*

$$P^{\bar{M}}_{success}(t) > P^{M}_{success}.$$

Then the biometric credential c_2 is asymptotically separable.

Proof. Assume by contradiction that the biometric credential is not asymptotically separable. Then, there exists $\varepsilon > 0$ such that for all t both $\frac{\mathit{FAR}(t)}{1-\mathit{FRR}(t)} \geq \varepsilon$ and $\frac{\mathit{FRR}(t)}{1-\mathit{FAR}(t)} \geq \varepsilon$. Since it is given that $\bar{M}$ uses both credentials c_1 and c_2 and achieves a strictly greater success probability than an atomic credential mechanism M, then there exists a threshold t such that either: $P^{\text{AND}}_{\text{success}}(t) > P^1_{\text{safe}}$ *or* $P^{\text{OR}}_{\text{success}}(t) > P^1_{\text{safe}}$. Thus, for every atomic credential c_1 necessarily either the inequality of AND mechanism holds (Equation 9) or the inequality of OR mechanism holds (Equation 11). In particular, these inequalities hold for an atomic credential with state probabilities $P^1_{\text{safe}}, P^1_{\text{leak}}, P^1_{\text{loss}}$ such that

$$\max\left\{\frac{P^1_{\text{leak}}}{P^1_{\text{safe}}}, \frac{P^1_{\text{loss}}}{P^1_{\text{safe}}}\right\} < \varepsilon.$$

Thus, for a mechanism $\bar{M}$ comprising this atomic credential it holds that either

$$\frac{\mathit{FRR}(t)}{1-\mathit{FAR}(t)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}} \cdot (1-\mathit{FAR}(t))} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}}}, \text{ or}$$
$$\frac{\mathit{FAR}(t)}{1-\mathit{FRR}(t)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}} \cdot (1-\mathit{FRR}(t))} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}}}.$$

However, by the contradiction assumption that the biometric credential is not asymptotically separable it follows that for every threshold t

$$\frac{\mathit{FRR}(t)}{1-\mathit{FAR}(t)} \geq \varepsilon > \frac{P^1_{\text{leak}}}{P^1_{\text{safe}}}, \text{ and}$$
$$\frac{\mathit{FAR}(t)}{1-\mathit{FRR}(t)} \geq \varepsilon > \frac{P^1_{\text{loss}}}{P^1_{\text{safe}}}.$$

This contradicts the fact that, for every atomic credential c_1, the success probability of $\bar{M}$ is strictly greater than that of M, as presented in both Equation 9 and Equation 11. Therefore, the biometric credential must be asymptotically separable. □

We showed that $\bar{M}$ achieves greater success probability than that of M the biometric credential is *asymptotically separable* and the atomic credential has $P^1_{\text{safe}} > 0$ and at least one of $P^1_{\text{leak}}, P^1_{\text{loss}} > 0$. We also presented that asymptotic separability is a necessary condition for $\bar{M}$ achieving greater success probability than every atomic credential mechanism. We extend our proof to a mechanism that uses a biometric credential and k atomic credentials and show that the mechanism achieves greater success probability than that of a mechanism using only the atomic credentials. First, every authentication mechanism M can be represented as a sum of products, i.e., a disjunctive normal form (DNF) [33]. Then, we define a *consolidated credential* of k atomic credentials as if it were itself a single atomic credential.

Definition 7. *Let M be an authentication mechanism over k atomic credentials. The* consolidated credential *of M is the single abstract credential whose availability to the user and attacker is determined by the outputs of M on their respective availability vectors $\sigma^U, \sigma^A \in \{0,1\}^k$. The fault probabilities of the consolidated credential are thus*

$$P^M_{safe} = \Pr\left[M(\sigma^U) = 1,\ M(\sigma^A) = 0\right],$$
$$P^M_{loss} = \Pr\left[M(\sigma^U) = 0,\ M(\sigma^A) = 0\right],$$
$$P^M_{leak} = \Pr\left[M(\sigma^U) = 1,\ M(\sigma^A) = 1\right],$$
$$P^M_{stolen} = \Pr\left[M(\sigma^U) = 0,\ M(\sigma^A) = 1\right].$$

Proposition 4. *Let $\{c_1, c_2, \ldots, c_k\}$ be a set of k independent atomic credentials, c_{k+1} be a biometric credential whose user distribution $f_U(t)$ and attacker distribution $f_A(t)$ are asymptotically separable and M be a mechanism using $\{c_1, c_2, \ldots, c_k\}$ whose consolidated credential has $P^M_{safe} > 0$ and at least one of $P^M_{leak}, P^M_{loss} > 0$. Then, there exists a mechanism $\bar{M}$ using $\{c_1, c_2, \ldots, c_k, c_{k+1}\}$ such that*

$$P^{\bar{M}}_{\text{success}} > P^{M}_{\text{success}}$$

Proof. Let M be a mechanism using $\{c_1, c_2, \ldots, c_k\}$. The mechanism M induces a reduction of the k credentials into a single consolidated credential. The consolidated credential of M is a single atomic credential with state probabilities $P^M_{\text{safe}} > 0$, and at least one of P^M_{leak}, or $P^M_{\text{loss}} > 0$. By Proposition 2, there exists a mechanism $\bar{M}$ using the consolidated credential and the biometric credential that achieves greater success probability than that of the consolidated credential mechanism. Since the consolidated credential is equivalent to the original mechanism M, it follows that there exists a mechanism $\bar{M}$ using $\{c_1, c_2, \ldots, c_k, c_{k+1}\}$ that achieves greater success probability than that of M. □

Proposition 5. *Let c_{k+1} be a biometric credential with user distribution $f_U(t)$ and attacker distribution $f_A(t)$. Suppose that for every set of k independent atomic credentials $\{c_1, \ldots, c_k\}$ and every mechanism M over these*

credentials whose consolidated credential has $P^M_{safe} > 0$ and at least one of $P^M_{leak}, P^M_{loss} > 0$, there exists a mechanism $\bar{M}$ using $\{c_1, \ldots, c_k, c_{k+1}\}$ such that

$$P^{\bar{M}}_{success}(t) > P^M_{success}.$$

Then the biometric credential c_{k+1} must be asymptotically separable.

Proof. The mechanism M induces a reduction of the k credentials into a single consolidated credential. The consolidated credential of M is a single atomic credential with state probabilities $P^M_{\text{safe}} > 0$, and at least one of $P^M_{\text{leak}}, P^M_{\text{loss}} > 0$. Then, $\bar{M}$ is a mechanism using the consolidated credential of M and the biometric credential. By Proposition 3, the biometric credential must be asymptotically separable. □

Since biometric credentials often exhibit a more common behavior [25], [31], we introduce the stronger *asymptotic dominance* condition to capture this property. Intuitively, asymptotic dominance means that one distribution function dominates the other in the tails of the distributions: for very low score values s the attacker distribution dominates the user distribution, and vice versa for very high values of s. We prove that a mechanism $\bar{M}$ that uses an *asymptotically dominated* biometric credential and an atomic credential achieves a greater success probability than that of a single atomic credential mechanism M, and show it is a sufficient condition (see Appendix A).

8. Authentication with Multiple Attempts

In biometric real world authentication, a mechanism commonly allows the principal to retry following failure [34], typically up to a bounded number of attempts before lockout [35]. Since retries change the mechanism and raise the success probability of users and attackers alike, the single attempt threshold is no longer optimal. We therefore formalize a mechanism that limits the number of attempts (§8.1). Within that limit, the mechanism success probability alone is not enough, since the threshold optimization may require many attempts to succeed. Then, we analyze both the mechanism's success probability and the mean number of attempts required for a user's first successful authentication (§8.2). As these two metrics conflict, we examine their tradeoff subject to different constraints (§8.3).

8.1. Predefined Numbers of Attempts

A deployed mechanism fixes its upper attempts number bound before any authentication takes place. We first consider a mechanism that allows at most k attempts, and show how changing the predefined attempts number k affects the mechanism success probability. The mechnanism may use a different threshold t_j at every attempt j, and the user fails to authenticate if he is rejected in all k attempts. Thus, the probability that the user is rejected at an arbitrary single attempt j is $\mathit{FRR}(t_j)$ and the probability that the user is accepted at least once, is the probability it's not rejected in all k attempts: $1 - \prod_{j=1}^{k} \mathit{FRR}(t_j)$. Mechanism success requires rejecting the attacker at every attempt. Since the attacker is rejected at each attempt j with probability $1-\mathit{FAR}(t_j)$, the probability that the attacker is rejected in all k attempts is: $\prod_{j=1}^{k} (1 - \mathit{FAR}(t_j))$. The authentication mechanism succeeds when the user is accepted at least once and the attacker is rejected in every attempt. Therefore, the overall success probability is

$$P^{(k)}_{\text{success}}(\mathbf{t}) = \left(1 - \prod_{j=1}^{k} \mathit{FRR}(t_j)\right) \prod_{j=1}^{k} (1 - \mathit{FAR}(t_j)). \tag{13}$$

Both the $\mathit{FRR}(t_j)$ and $\mathit{FAR}(t_j)$ error rates are, by definition, lower or equal than 1, for every attempt j and threshold t_j. The products of $\mathit{FRR}(t_j)$ and the products of $1-\mathit{FAR}(t_j)$ are both monotonically decreasing functions with k. As a result, for a fixed thresholds selection, increasing the attempts number k increases the user probability to be accepted, while reducing the probability of the attacker to be rejected. Increasing k can therefore help the user gain access, but also increases the risk of accepting an attacker. The thresholds selection, when optimizing the success probability, must therefore balance these two effects.

8.2. User Authentication Within k Attempts

We are now interested not only in whether the user eventually authenticates, but also in how many attempts it takes for a user successful authentication. We proceed in three steps. First, we derive the probability that the user's first successful authentication occurs at a given attempt, which yields the probability of authenticating within k attempts. Then, we consider the conditional attempt number given that the user successfully authenticates. Finally, we take the mean of this conditional distribution, obtaining the average number of attempts a user requires. Let N_U denote the attempt number at which the user authenticates for the first time. If the mechanism rejects the user in all k attempts, we define $N_U = \infty$. The event $\{N_U = j\}$ means that the user is rejected in attempts $1, \ldots, j-1$ and is then accepted at attempt j. Under the assumption that every authentication attempt is independent, the probability that the user authenticates for the first time at attempt j is therefore

$$\Pr[N_U = j] = \left(\prod_{i=1}^{j-1} \mathit{FRR}(t_i)\right) (1 - \mathit{FRR}(t_j)). \tag{14}$$

This is the *first success probability* that the user successfully authenticates specifically at attempt j, having failed every preceding attempt. Summing the first success probabilities over all permitted attempts gives the probability that the user successfully authenticates within k attempts:

$$\Pr[N_U \leq k] = \sum_{j=1}^{k} \Pr[N_U = j].$$

Substituting the first success probabilities yields

$$\Pr[N_U \leq k] = \sum_{j=1}^{k} \left(\prod_{i=1}^{j-1} \mathit{FRR}(t_i) \right) (1 - \mathit{FRR}(t_j)) .$$

As described in the previous section, this probability equals

$$\Pr[N_U \leq k] = 1 - \prod_{i=1}^{k} \mathit{FRR}(t_i). \tag{15}$$

Having derived the probability that the user successfully authenticates within k attempts, we now consider the conditional distribution of the attempt number N_U given that the user successfully authenticates within k attempts. By the definition of conditional probability,

$$\Pr[N_U = j \mid N_U \leq k] = \frac{\Pr[\{N_U = j\} \cap \{N_U \leq k\}]}{\Pr[N_U \leq k]}.$$

For every $j \in \{1, \ldots, k\}$, the event $\{N_U = j\}$ implies that the user succeeds within k attempts. In set notation,

$$\{N_U = j\} \subseteq \{N_U \leq k\}.$$

Consequently, their intersection is simply

$$\{N_U = j\} \cap \{N_U \leq k\} = \{N_U = j\}.$$

Then it follows that

$$\Pr[N_U = j \mid N_U \leq k] = \frac{\Pr[N_U = j]}{\Pr[N_U \leq k]}.$$

Using Equation 14 and Equation 15, we obtain

$$\Pr[N_U = j \mid N_U \leq k] = \frac{\left(\prod_{i=1}^{j-1} \mathit{FRR}(t_i) \right) (1 - \mathit{FRR}(t_j))}{1 - \prod_{i=1}^{k} \mathit{FRR}(t_i)}.$$

The numerator is the probability that the first successful authentication occurs exactly at attempt j. The denominator is the probability that the user succeeds at some attempt no later than k. Their ratio therefore gives the fraction of successful users whose first successful authentication occurs at attempt j.

The conditional mean number of attempts is the expected value of N_U among users who successfully authenticate within k attempts. By the definition of conditional expectation for a discrete random variable,

$$\mathbb{E}\left[N_U \mid N_U \leq k\right] = \sum_{j=1}^{k} j \Pr[N_U = j \mid N_U \leq k].$$

Substituting the conditional probabilities gives

$$\mathbb{E}\left[N_U \mid N_U \leq k\right] = \frac{\displaystyle\sum_{j=1}^{k} j \left(\prod_{i=1}^{j-1} \mathit{FRR}(t_i) \right) (1 - \mathit{FRR}(t_j))}{1 - \displaystyle\prod_{j=1}^{k} \mathit{FRR}(t_j)}.$$

Conditioning on $N_U \leq k$ is necessary because $N_U = \infty$ when the user is rejected in all permitted attempts. The conditional mean measures the average number of attempts required specifically among successful authentication sessions, helping system designers to determine their requirements and priorities when optimizing thresholds over attempts.

8.3. Optimization Under Mean-Success Trade-Off

System designers evaluate the security of a multiple attempt authentication mechanism by two metrics: (1) the conditional mean number of a user authentication attempts, and (2) the overall success probability of the mechanism. The first metric measures usability, while the second metric measures reliability. However, there is an inherent trade-off between these metrics. As the number of attempts grows, the overall success probability increases while usability degrades.

We consider two optimization problems: one that minimizes the conditional mean number of user authentication attempts subject to a constraint on the success probability, and another that maximizes the success probability subject to a constraint on the conditional mean number of user authentication attempts.

We start by minimizing the conditional mean number of user authentication attempts. Thus

$$\begin{aligned} \underset{\mathbf{t}}{\text{minimize}} \quad & \mathbb{E}\left[N_U \mid N_U \leq k\right] \\ \text{subject to} \quad & P_{\text{success}}(\mathbf{t}) \geq P'. \end{aligned}$$

Equivalently, the constraint is

$$\left(1 - \prod_{j=1}^{k} \mathit{FRR}(t_j) \right) \prod_{j=1}^{k} (1 - \mathit{FAR}(t_j)) \geq P'.$$

Without the constraint, this problem has a trivial solution: set every threshold to accept everything, and the user always succeeds on the first attempt. The mean is then 1, but so is the attacker's success rate. The constraint P' rules this out. At the optimum, the constraint holds with equality. Any solution with a strictly higher success probability could trade that slack for a more permissive threshold and a lower mean. Thus, this optimization enforces a minimum security level while maximizing the usability within it.

We next maximize the mechanism's success probability subject to an upper bound on the conditional mean number of user authentication attempts. Let μ' denote this upper bound. For fixed attempt limits k, we optimize the threshold vector $\mathbf{t} = (t_1, \ldots, t_k)$. Thus

$$\begin{aligned} \underset{\mathbf{t}}{\text{maximize}} \quad & P_{\text{success}}(\mathbf{t}) \\ \text{subject to} \quad & \mathbb{E}\left[N_U \mid N_U \leq k\right] \leq \mu'. \end{aligned}$$

Substituting the success probability and the conditional mean gives

$$\underset{\mathbf{t}}{\text{maximize}} \quad \left(1 - \prod_{j=1}^{k} \mathit{FRR}(t_j)\right) \prod_{j=1}^{k} (1 - \mathit{FAR}(t_j))$$

$$\text{subject to} \quad \frac{\sum_{j=1}^{k} j \left(\prod_{i=1}^{j-1} \mathit{FRR}(t_i)\right) (1 - \mathit{FRR}(t_j))}{1 - \prod_{j=1}^{k} \mathit{FRR}(t_j)} \leq \mu'.$$

We restrict the optimization to threshold vectors for which $P_U(\mathbf{t}) > 0$, so that the conditional mean is defined. This optimization maximizes the mechanism's success probability, while ensuring that successful user authentication sessions require at most μ' attempts on average.

Here the constraint plays a different role. Without the constraint, the optimization balances both the *FRR* and *FAR*, leading to a non trivial solution as described in (§8.1). The constraint μ' caps how long a successful user waits. Unlike the previous problem, this constraint may not bind at all: if the optimal threshold vector already lets users authenticate within μ' attempts on average, nothing changes. When it does bind, the optimization enforces thresholds more permissive than the unconstrained problem, leading to a reduction in the overall success probability. Thus, this optimization fixes a usability budget and maximizes the mechanism's success probability within it.

8.4. Spoofing Attack With Multiple Attempts

We simulate a spoofing attack scenario in which a principal has multiple attempts to authenticate. We use the same genuine and impostor distributions calculated in the spoofing attack scenario using the LivDet2015 database.

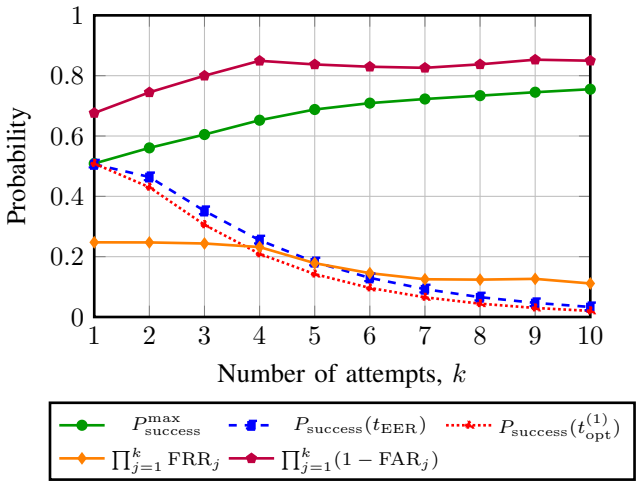
We follow the theoretical analysis and calculate the success probability under multiple attempts of a mechanism that uses a biometric credential only (Equation 13). We then simulate several authentication attempts by a principal P. For each number of attempts k, we calculate the mechanism's success probability as the probability that the user is accepted at least once, while the attacker is rejected on all k attempts. As the success probability depends on the number of attempts k, we compare the mechanism's success probability under three thresholds selections baselines: (1) the *EER* threshold reused for all k attempts, (2) the single-attempt optimal threshold reused for all k attempts, and (3) threshold optimization over all k attempts, as shown in Figure 20. The first two baselines represent biometric authentication systems that use a fixed threshold across multiple authentication attempts, either the *EER* threshold or the single-attempt optimal threshold, without optimizing it for multiple attempts.

We find that the success probability of this mechanism, which uses a biometric credential only, increases with the number of attempts k and converges to 1 when optimizing the threshold over all k attempts (Figure 21). The optimization suggests that the threshold should vary with the total number of attempts k. Furthermore, our findings reveal that reusing a single threshold in all k attempts, either the *EER* threshold or the single-attempt optimal threshold, leads to a reduction of up to 75% in the mechanism's failure probability. This reduction indicates that the attacker is more likely to succeed in spoofing the biometric credential as the number of attempts increases.

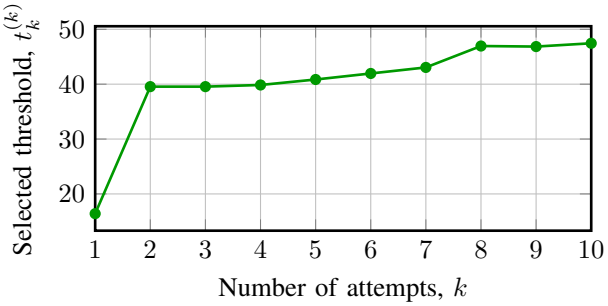
Previously, we noted that for each number of attempts k, the optimal threshold varies with k. Thus, we investigate how the optimal threshold selections are affected when we constrain the mean number of attempts, seeking to maximize the mechanism's success probability. We find that the optimal threshold increases with the number of attempts k, and that the mechanism's success probability increases as the mean number of attempts increases. Nevertheless, system designers seek to minimize the mean number of a user's authentication attempts.

We explore the tradeoff between the mean number of attempts and the mechanism's success probability. First, we identify the optimal threshold selections that minimize the mean number of user authentication attempts, while requiring the mechanism's success probability to exceed $P_{\text{success}}^{\min}$. This analysis shows that the greater the required success probability $P_{\text{success}}^{\min}$, the greater the mean number of user authentication attempts $\mathbb{E}[N_U]$ required to achieve it. Moreover, as presented in the theoretical analysis, the mechanism's success probability is equal to the success constraint. In addition, our findings reveal that a higher success probability constraint leads to a greater cumulative user success probability and a lower cumulative attacker success probability. This behavior indicates that the selected thresholds tend to be stricter in the initial attempts, with smaller differences between successive thresholds as the attempt index increases. In this way, the mechanism limits the probability of a successful spoofing attack, while relying on the user's cumulative success probability to increase with the number of attempts. These findings are summarized in Figure 22.

We also analyze the reverse direction, in which a system designer seeks to maximize the achievable security under a usability constraint. Specifically, we constrain the mean number of user authentication attempts $\mathbb{E}[N_U]$, and seek to maximize the mechanism's success probability. We find that allowing a greater mean number of user authentication attempts $\mathbb{E}[N_U]$, leads to a lower cumulative success probability for the attacker. A higher limit on the mean number of attempts leads to stricter threshold selections in the initial authentication attempts. In this way, the mechanism prioritizes preventing the attacker from spoofing the biometric credential, over accepting the user in the initial attempts. Thus, the user's cumulative authentication probability is initially lower than under a lower limit on the mean number of attempts. However, it increases significantly over subsequent attempts, as the user has more opportunities to obtain a score above the threshold. These findings are shown in Figure 23.



(a) Cumulative success and rejection probabilities.



(b) Threshold applied at the final attempt of each policy.

Figure 20: Success probabilities and selected thresholds as a function of the maximum number of authentication attempts k. (a) The optimized cumulative success probability, compared with reusing the *EER* threshold and the single-attempt optimal threshold, together with the products of user-rejection and attacker-rejection probabilities. (b) The threshold applied at attempt k of the corresponding k-attempt fixed-prefix policy. Each value of k represents a separately optimized policy.

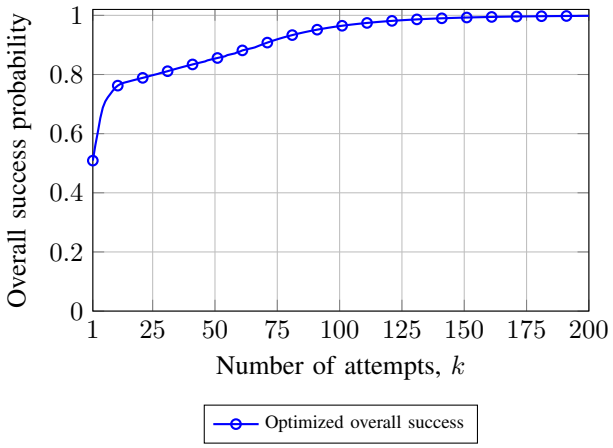


Figure 21: Optimized mechanism success probability as a function of the number of authentication attempts k. The threshold policy is optimized separately for each k, without constraints on the conditional mean number of attempts or the success probability. The dashed line marks probability one.

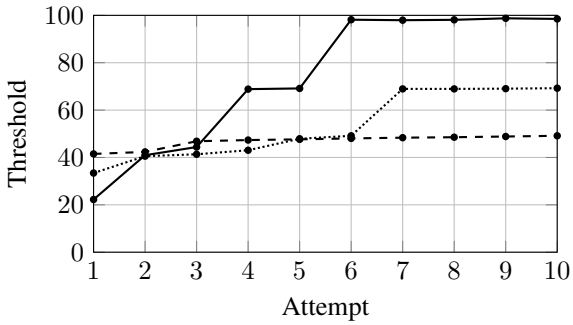
9. Biometric Sensors Survey

We assess whether biometric sensors allow tuning the match threshold. We review vendors' specifications of commercially available biometric sensors, spanning fingerprints, face, iris and palm. We identify 32 device families from 21 manufacturers and classify each family according to two criteria that reflect our model's requirements.

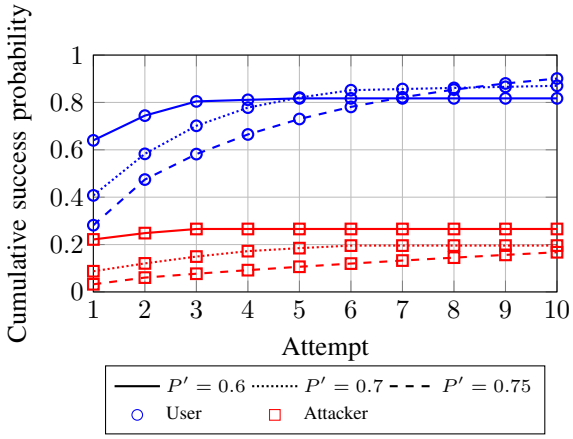
The first criterion is *in-sensor decision*: whether the device performs matching and emits an accept/reject verdict itself, or merely captures a sample for a host matcher. Of the 32 families, 23 make an in-sensor decision while the remaining 9 are capture only. A capture-only sensor exposes either the biometric data, matching score or the sample itself. Revealing the biometric data or matching score can facilitate hill-climbing attacks, in which an attacker repeatedly modifies a biometric sample based on score feedback until the system accepts it [36]. Furthermore, exposing the biometric samples creates privacy risks, as they may be reused to identify users across services without their knowledge or consent [37]. For these reasons, we exclude these 9 families and focus on sensors that expose only an accept/reject decision.

The second criterion is *threshold tuning*: whether the device allows the integrator to set a numeric match threshold. We distinguish this capability from three weaker methods, none of which permits setting a specific match threshold. The first method tunes a *target error rate*, where the integrator specifies a desired *FAR* or *FRR* value within a predetermined range. The error rates the integrator requires may lie outside that range entirely. Moreover, since *FAR* and *FRR* are functions of the match threshold, specifying a target error rate does not allow the integrator to determine the corresponding threshold, unless these functions are known. Instead, the device selects the threshold internally without exposing its value. The second method allows choosing a *discrete security level* such as 1-5, 1-9, High/Medium/Low or selected *FAR* discrete values. The integrator is therefore limited to a few fixed levels that influence the match threshold without setting it directly, and the optimal threshold may not be among them. Moreover, even when a level is labelled with its error rate, the corresponding threshold value behind it is not exposed, so no match threshold outside the vendor's set can be reached. The third method exposes no threshold parameter of any kind, leaving the integrator with the single match threshold the vendor chose.

The third criterion is *error-rate exposure*: whether the vendor publishes the *FAR* and *FRR* for the device. Setting a threshold is necessary but not sufficient for optimization. Without knowing the *FAR* and *FRR* functions, the integrator

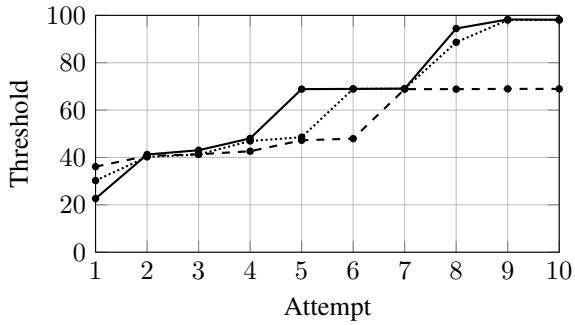


(a) Selected thresholds.

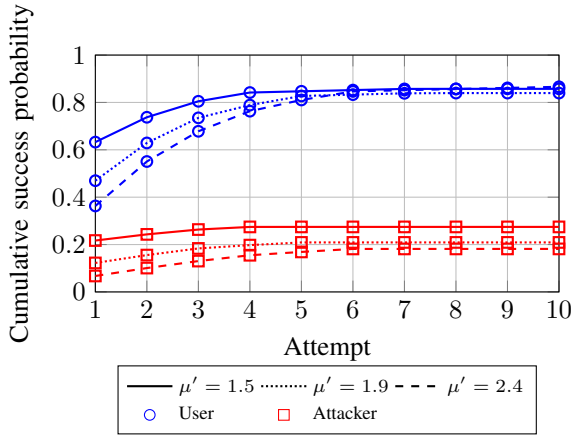


(b) Cumulative user and attacker acceptance probabilities.

Figure 22: Policies minimizing the conditional mean number of user authentication attempts subject to $P_{\text{success}} \geq P'$, with $k = 10$. The upper panel shows selected thresholds; the lower panel shows the probability of at least one user or attacker acceptance through each attempt. Line styles identify constraint values, while blue circles represent the user and red squares represent the attacker. The constraint applies to the complete policy.



(a) Selected thresholds.



(b) Cumulative user and attacker acceptance probabilities.

Figure 23: Policies maximizing the mechanism's success probability subject to $\mathbb{E}[N_U \mid N_U \leq k] \leq \mu'$, with $k = 10$. The upper panel shows selected thresholds; the lower panel shows the probability of at least one user or attacker acceptance through each attempt. Line styles identify constraint values, while blue circles represent the user and red squares represent the attacker. The constraint applies to the complete policy.

can set t to a specific value, but cannot determine the error rates it induces.

Only 2 out of the 23 in-sensor decision families satisfy the threshold tuning criterion. Those families, comprising 6 individual models, permit setting an arbitrary numeric match threshold. None of the 23 satisfies the error-rate exposure criterion: no vendor publishes *FAR* or *FRR* functions, so an integrator that can set a threshold still cannot determine which value to set. The detailed list is in Appendix B.

The practical consequence is that today's authentication mechanisms are built atop in-sensor decision devices and cannot, in any of the 23 families we surveyed, optimize the match threshold. In 21 of the 23 families we surveyed the threshold cannot be set at all, and in the remaining two it can be set but not informed, since the error-rate functions are not published. To support threshold optimization, vendors should expose both the match threshold and the *FAR* and *FRR* functions that characterize it. This would allow integrators to select thresholds according to their requirements, and as part of the other credentials used in the mechanism.

10. Conclusion

In a single-attempt mechanism using a biometric credential alone, optimizing the threshold yields only small improvements, which may explain why it has received limited attention. But when the credential is combined with other credentials or used across multiple attempts, the optimal threshold depends on the mechanism and differs from the heuristic one. We characterize a condition under which adding a biometric credential always improves the mechanism's success, and prove it. We also find that the optimal threshold depends on the number of allowed attempts, and present two constrained optimization problems that let system designers balance security and usability. Simulating spoofing attacks on public biometric datasets, we find that

threshold optimization reduces the failure probability by 20-72% compared to the heuristic threshold, and by up to 21% compared to using the atomic credential alone. Reusing the heuristic threshold across attempts reduces the success probability by up to 72 percentage points.

However, our survey of 32 commercial device families from 21 manufacturers finds that only two allow setting a numeric match threshold, and none publishes the *FAR* and *FRR* functions required to choose one. We therefore propose that vendors should expose both, so integrators can select thresholds for their mechanisms. This paves the way for developing new authentication protocols based on threshold selection and may enhance the security of digital and physical assets for millions of users around the world.

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Appendix A.
A Sufficient Condition for Integrating Biometric Credentials

Since biometric credentials often exhibit a more common behavior [25], [31], we introduce an *asymptotic dominancy* condition which captures this property. We prove that adding an *asymptotically dominated* biometric credential can produce a new mechanism $\bar{M}$ with higher success probability than that of a single atomic credential mechanism M, and show it is a sufficient condition.

Definition 8. *A biometric credential with a user distribution function $f_U(s)$ and an attacker distribution function $f_A(s)$ are* asymptotically dominated *if*

$$\forall c > 0\ \exists t \text{ s.t. } : \forall s \geq t : f_A(s) \leq c\, f_U(s),\ \textit{or}$$
$$\forall c > 0\ \exists t \text{ s.t. } : \forall s \leq t : f_U(s) \leq c\, f_A(s).$$

Proposition 6. *Let c_1 be an atomic credential with $P^1_{safe} > 0$ and at least one of $P^1_{leak}, P^1_{loss} > 0$, c_2 be a biometric credential whose user distribution $f_U(t)$ and attacker distribution $f_A(t)$ are asymptotically dominated and M be a mechanism using c_1. Then, there exists a mechanism $\bar{M}$ using c_1, c_2, such that*

$$P^{\bar{M}}_{\text{success}} > P^{M}_{\text{success}}$$

Proof. The success probability of a mechanism comprising an atomic credential only is P^1_{safe} (Equation 4). A biometric credential is characterized by four state probabilities that are functions of *FAR* and *FRR*

$$\begin{aligned}
P^2_{\text{safe}}(t) &= 1 - \mathit{FAR}(t) - \mathit{FRR}(t) + \mathit{FAR}(t) \cdot \mathit{FRR}(t),\\
P^2_{\text{loss}}(t) &= \mathit{FRR}(t)\big(1 - \mathit{FAR}(t)\big),\\
P^2_{\text{leak}}(t) &= \mathit{FAR}(t)\big(1 - \mathit{FRR}(t)\big),\\
P^2_{\text{theft}}(t) &= \mathit{FAR}(t) \cdot \mathit{FRR}(t).
\end{aligned}$$

We previously observed that a sufficient condition for producing a new mechanism with greater success is described by Equation 10 and Equation 12. Thus, we need to show that there exists an operating threshold t such that either

$$\frac{\mathit{FRR}(t)}{1 - \mathit{FAR}(t)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}}},\ \textit{or}$$
$$\frac{\mathit{FAR}(t)}{1 - \mathit{FRR}(t)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}}}$$

We expand left hand side of Equation 10.

$$\frac{\mathit{FRR}(t)}{1 - \mathit{FAR}(t)} = \frac{\int_{-\infty}^{t} f_U(s)ds}{\int_{-\infty}^{t} f_A(s)ds}$$

By Definition 8, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold t such that for every $s \leq t$, $f_U(s) \leq c \cdot\ f_A(s)$. Therefore:

$$\frac{\int_{-\infty}^{t} f_U(s)ds}{\int_{-\infty}^{t} f_A(s)ds} \leq \frac{\int_{-\infty}^{t} c \cdot f_A(s)ds}{\int_{-\infty}^{t} f_A(s)ds} = c$$

This holds $\forall c > 0$,; thus, we can choose c to equal the right hand side of Equation 10 and satisfy the sufficient condition. Similarly, we expand left hand side of Equation 12.

$$\frac{\mathit{FAR}(t)}{1 - \mathit{FRR}(t)} = \frac{\int_{t}^{\infty} f_A(s)ds}{\int_{t}^{\infty} f_U(s)ds}$$

By Definition 8, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold t such that for every $s \geq t$, $f_A(s) \leq c \cdot\ f_U(s)$. Therefore,

$$\frac{\int_{t}^{\infty} f_A(s)ds}{\int_{t}^{\infty} f_U(s)ds} \leq \frac{\int_{t}^{\infty} c \cdot f_U(s)ds}{\int_{t}^{\infty} f_U(s)ds} = c, \forall c > 0$$

We have showed that $\forall c > 0$ either $\frac{\mathit{FAR}(t)}{1-\mathit{FRR}(t)} < c$ or $\frac{\mathit{FRR}(t)}{1-\mathit{FAR}(t)} < c$. Therefore, integrating an asymptotically

dominated biometric credential with an atomic credential produces $\bar{M}$ with greater success probability. □

This proof is non trivial, as the integration of an atomic credential mechanism with another atomic credential, does not necessarily produce a new mechanism with higher success probability. Moreover, we have shown that integrating a biometric credential into an atomic credential mechanism without adjusting the match threshold t may reduce the mechanism's success probability $(\S 6.2)$. Thus, Proposition 2 and Proposition 6 emphasize the importance of adjusting the threshold when adding a biometric credential into an authentication mechanism. The *asymptotically dominated* condition reflects practical scenarios, as for very low score values s the attacker distribution dominates the user distribution, and vice versa for very high values of s. We validate this claim with real public biometric data in Section **??**.

Appendix B. Biometric Sensors Survey Table

Table 3 shows all the commercially available devices we surveyed. Rows are grouped by threshold exposure, from devices permitting an arbitrary match threshold down to those exposing none, followed by the capture-only devices outside our scope. The *On-sensor* column indicates whether the device performs matching and emits an accept/reject verdict itself. The *Threshold exposure* column gives the mechanism, if any, by which an integrator may influence the decision boundary. Families from one manufacturer sharing a modality and threshold mechanism are listed on a single row, so the family counts in the group headings exceed the number of rows. Each entry cites the vendor documentation from which its classification was drawn.

TABLE 3: Surveyed biometric device families, grouped by threshold exposure. On-sensor column indicates the device performs matching and emits an accept/reject verdict itself. Families from one manufacturer sharing a modality and threshold mechanism are listed on a single row.

Device family	Manufacturer	Modality	On-sensor	Threshold exposure	Ref.
Arbitrary match threshold (2 families, 6 models)					
F18, SpeedFace-V5L, MultiBio700, SF100, F21	ZKTeco	FP / face / palm	Yes	Numeric score	[38], [39], [40]
DS-K1T342 series	Hikvision	Face / FP	Yes[†]	Numeric score	[41], [42]
Target error rate (2 families)					
EF-45, EF-45N, EF-45NC, EF-45 AC	CMITech	Iris / face	Yes	FMR 10^{-5}–10^{-14}	[43]
MorphoAccess SIGMA Lite/Lite+/Wide/Extreme	IDEMIA	FP / face	Yes	FAR 1%–10^{-7}%	[44], [45]
Discrete security level (12 families)					
R307 / AS608, R503, R502-A	Hangzhou Grow	FP	Yes	Level 1–5	[46], [47], [48]
ID809 (SEN0348 / SEN0359)	DFRobot	FP	Yes	Level 1–5	[49]
NAC-3000, Fingkey Access	NITGEN	FP	Yes	Level 1–9, per mode	[50]
FIM5360 / FIM6060 / FIM5260 / FIM4060	NITGEN	FP	Yes	Level, per user	[51]
AC-7000	Union Community	Face / FP	Yes	Level 1–9, per mode	[52]
ID F450 / F455	RealSense	Face	Yes	High/Med/Low	[53]
BioStation 3, FaceStation F2	Suprema	Face / FP	Yes	Level, not enumerated	[54], [55]
SFM Slim, SFM5500	Suprema	FP	Yes	FAR values, per user	[56], [57]
No threshold parameter (7 families)					
GT-521F32 / GT-521F52	ADH-Tech	FP	Yes	None	[58]
BM-Lite (SafeTouch)	Fingerprint Cards	FP	Yes	None	[59]
IriShield MK 2120U / UL	IriTech	Iris	Yes	Not documented	[60]
FVA-U4ST	mofiria	Finger vein	Yes	Not documented	[61]
iCAM7000 series	Iris ID	Iris	Mode-dep.[‡]	Not documented	[62]
PalmSecure ID Match	Fujitsu	Palm vein	Yes	Not documented	[63]
IXM TITAN / TOUCH 2 / TFACE	Invixium	Face / FP / vein	Yes	Unverified	[64]
Capture only — outside scope (9 families)					
DigitalPersona 4500	HID Global	FP	No	Host: numeric	[65], [66]
Hamster Pro 20 (HU20)	SecuGen	FP	No	Host: score 0–199	[67], [68]
BioMini Slim 2	Suprema	FP	No	Host SDK	[69]
ZK9500 / SLK20R	ZKTeco	FP	No	Host SDK	[70]
MFS100	Mantra Softech	FP	No	Host SDK / RD service	[71]
PalmSecure-F Pro+	Fujitsu	Palm vein	No	Host library	[72]
H-1 (PC-KCA100/110)	Hitachi	Finger vein	No	Host engine	[73]
BMT-20	CMITech	Iris	No	Host SDK	[74]
SEN0626	DFRobot	Face	No matcher	n/a[§]	[75]

[†]Self-deciding only in Terminal Authentication Mode; in Reader Authentication Mode the decision is delegated to an upstream controller. [‡]On-device matching is one of three selectable operating modes. [§]Face detection only: no template store and no identity decision.